Discrete Math Class Notes

These are class notes for a first year undergrad Discrete Math class taught from Grimaldi's 'Discrete and Combinatorial Mathematics (5th edition)', which we will refer to as 'the text'.

1 Counting

This section draws from Sections 1.1 - 1.4 of the text. It is understood that the students have seen basic counting techniques, so we only review these topics quickly. Many of what would be examples are posed as problems.

Topics Covered:

- Rule of Sum
- Rule of Product
- Counting the Complement
- Permutations of sets and multisets
- Circular permutations
- Combinations
- Combinations with replacement

A *set* is a collection of objects, which we call *elements* or *members* of the set. For the time being, the sets we consider will be finite sets in which no member occurs more than once.

Example 1.1. The collection S of two digit integers not divisible by 3 is a set:

$$S = \{10, 11, 13, 14, 16, 17, 19, \ldots, 97, 98\}$$

We are interested in counting $|S|$, the number of elements in S. Clearly we could just count, and find that there are 60. There are some basic techniques that allow us to simplify this task.

1.1 Basic Counting Techniques:

The Rule of Sum (Divide and Conquer): If we can break a set A into smaller sets $A_1, \ldots, A_d$ such that every element is in exactly one set, then

$$|A| = |A_1| + \cdots + |A_d|.$$

For example, we could break the set S from Example 1.1 up into the set $S_1 = \{10, 13, 16, 19, \ldots, 94, 97\}$ of the form $3x + 1$ for some integer x, and the set $S_2 = \{11, 14, 17, 20, \ldots, 95, 98\}$ of the form $3x + 2$. It is not hard to see that $|S_1| = |S_2|$ so we only have to count one of them to see that $|S| = |S_1| + |S_2| = 30 + 30 = 60$.

The Rule of Product: One good way of counting elements of a set is considering the process of choosing an element of the set. If the process can be broken down into independent steps, we can simply cound the ways of completing each step.

For example, the elements of S are integers of the form $3x + y$ where x is in $\{3, 4, \ldots, 32\}$ and y is in $\{1, 2\}$. So we construct every element by choosing an x and a y. We can choose x in 30 ways and y in 2 ways, so there are $30 \cdot 2 = 60$ ways of choosing an element of S.

Counting the Complement: Sometimes it is easy to count a bigger set and then remove items not in the wanted set.

For example, it is easier to count the set T of integers between 1 and 99 that are not divisible by 3. There are 66 of them. The *complement* $\overline{S}$ of S in T, that is the set of elements of T not in S is $\overline{S} = \{1, 2, 4, 5, 7, 8\}$, so $|S| = |T| - |\overline{S}| = 66 - 6 = 60$.

1.2 Permutations and Combinations

1.2.1 Permutations of a set

A *permutation* of a set is an arrangement of the items in a line. So $1, 2, 3$ and $3, 1, 2$ are different permutations of the set $\{1, 2, 3\}$. Using the rule of product, one can show that there are

$$n! = n \cdot (n - 1) \cdot \ldots \cdot 1$$

permutations of an n element set. Indeed, there are n ways to choose the first element, and then $n - 1$ ways to choose the second, etc. The notation '$n!$' is read 'n factorial'.

An r-*permutation* or a *size-r permutation* of set is an arrangement of r items of the set in a line. Again by the rule of product, one can show that there are

$$P(n,r) := \frac{n!}{(n-r)!}$$

r-permutations of an n element set.

1.2.2 Permutations of a multiset

A *multiset* is a collection of elements that are not necessarily distinct. The multiset $\{1,2,2,2\}$ is different from $\{1,2\}$, though as sets, they would be the same. Counting permutations of a multiset adds a slight complication to the task of counting permutations of a set: certain permutations are indistinguishible, so we consider them the same. The multiset $\{1,2,2,2\}$ only has 4 distinguishible permutations:

$$\{(1,2,2,2),(2,1,2,2),(2,2,1,2),(2,2,2,1)\}.$$

This is less than the $4! = 24$ permutations of a 4-element set.

We could get this by counting the elements of the multiset as distinct, in $4!$ ways, and then observing that each distinguishible permutation of the multiset is counted $3!$ times, one for each permutation of the repeated elements. So there are $4 = 4!/3!$ distinguisible permutations of the multiset. More generally we have...

Rule: In a multiset containing n_i *indistinguisible* elements of type i for $i = 1,\ldots,r$, the number of *distinguishible* permutations is

$$\frac{(n_1+n_2+\cdots+n_r)!}{n_1!\cdot n_2!\cdot\cdots\cdot n_r!}.$$

When we talk of permutations of a multiset, it is understood that we are talking of distinguishible permutations, so we will generally drop the 'distinguishible.'

Problem 1.1. Find the number of permutations of the letters MISSISSIPPI.

Problem 1.2. Find the number of permutations of the letters ABCDEFGH in which the letters of the word CAGE occur in this order. In which the word CAGE doesn't occur.

Problem 1.3. Count the number of paths in the plane from $(0,0)$ to $(4,9)$, using only steps that go one unit up or 1 unit to the left

Hint

Text Example 1.14

A *circular permutation* of a set is an arrangement of the elements in a circle. Two such arrangements are considered the same if you can get from one to the other by rotating the circle. These are the permutations we are considering in the next problem.

Problem 1.4.

(a) How many ways can 6 people be seated around a circular table. (That is, how many circular permutations are there of a 6 element set?)

(b) How many ways can 3 men and 3 women be seated around a circular table so that men and women alternate.

(c) How many ways can 3 couples be seated around a table so that the couples are seated together.

1.3 Combinations without replacement

An r-combination of an n-element set is a choice of r elements from the set. If the r-elements must be distinct, there are

$$\binom{n}{r} = P(n,r)/r! = \frac{n!}{(n-r)!r!}$$

of them, (They are permutations in which order is disregarded.) The notation $\binom{n}{r}$ is read 'n choose r', and is called a *binomial co-efficient* because of the following theorem.

Theorem 1.2 (Binomial Theorem)**.**

$$(x+y)^n = \sum_{i=0}^{n} \binom{n}{i} x^{n-i} y^i$$

Proof. Indeed, $(x+y)^n$ is the product of n factors $(x+y)$. Every summand in the expansion of $(x+y)^n$ corresponds to a choice of either x or y from each of these terms. The summand $x^{n-i}y^i$ will occur once for every way we can choose i of the n terms to choose y from. This is $\binom{n}{i}$ ways. □

Here is the typical problem.

Example 1.3. A pizza place has 16 different toppings. On a deluxe pizza you can choose 6 different toppings. How many different deluxe pizzas are there?

Solution

This is pretty straight forward: $\binom{16}{6}$.

See if you can deal with the following complications of this problem.

Problem 1.5. A pizza place offers the following toppings: 4 kinds of cheese, 7 kinds of vegetables, and 5 kinds of meat. How many different pizzas have

i. 2 cheeses, 4 vegetables and 3 meats (all different).

ii. 6 different toppings at least 3 of which are meat.

iii. 6 different toppings at least 3 of which are cheese.

Some problems can be dealt with as permutations or combination problems.

Example 1.4. How many ways can we split 20 people into 4 teams A, B, C, D of 5 people each. (Teams are distinguishable.)

Solution

We can view this as a permutation problem or a combination problem. We can choose 5 for the first team, 5 for the second, etc., so there are

$$\binom{20}{5}\binom{15}{5}\binom{10}{5}\binom{5}{5}.$$

Alternately, we can arrange 5 of each of A, B, C, D among 20 people:

$$\frac{20!}{5!5!5!5!}$$

You can show that these quantities are the same.

Example 1.5. How many ways can you arrange the letters MISSISSIPPI such that there are no adjacent I's.

Solution

We can do this using combinations and permutations together. First arrange the letters MSSSSPP in $7!/(4!2!)$ ways. Given such an arrangement: SMSPSSP, there are 8 spaces in which we can put one I: -S-M-S-P-S-S-P-. There are $\binom{8}{3}$ ways to place the 3 I's in these spaces, so the total number of arrangements is

$$\frac{7!}{4!2!}\binom{8}{3}.$$

Problem 1.6. What is the co-efficient of x^3y^3 in

i. $(x+y)^6$?

ii. $(2x+y)^6$?

1.4 Combinations with replacement

Consider the following question:

Example 1.6. A doughnut store has kruellers, fritters and honey glazed doughnuts. How many ways can you select a dozen?

Observe that there is a one-to-one correspondence between the ways to select a dozen, and the arrangements of 12 stars: '$*$' and 2 dividers: '$|$'. The arrangement

$$** \,|\, * * * * * * * \,|\, * * *$$

partitions the 12 stars into three parts. This corresponds to the choice of 2 kruellers, 7 fritters, and 3 honey glazed. Thus there are $\binom{12+2}{2}$ choices of the dozen. It is clear that this also counts the solutions of

$$x_1 + x_2 + x_3 = 12$$

where x_1, x_2, x_3 are non-negative integers. The value of x_i in a solution corresponds to the number of '$*$'s in the ith part.

Here are a couple of variations on the question.

Problem 1.7. How many solutions are there to the equation $x_1 + x_2 + x_3 = 12$ where all x_i are non-negative integers and $x_2 \geq 2$?

Hint

(1.7) One can do this by counting the non-negative integer solutions of $x_1 + x_2' + x_3 = 10$ (where $x_2' = x_2 - 2$).

Problem 1.8. How many solutions are there to the equation $x_1 + x_2 + x_3 \leq 12$ where all x_i are non-negative integers.

Problem 1.9. A composition of an integer n is an ordered sum of positive integers totalling to n. (So 8, $7+1$, $1+7$ and $1+1+1+1+1+1+1+1$ are all different compositions of 8.) How many compositions are there of 8?

Hint

(1.8) One can do this by counting the non-negative integer solutions of $x_1 + x_2 + x_3 + x_4 = 12$, (where x_4 is called a slack variable.)

Problem 1.10. How many ways can seven 'H's and nine 'T's be arranged in a line so that there exists exactly four strings of consecutive 'T's?

2 Symbolic Logic

This draws from Chapter 2 of the text.

Recall that an integer is prime if it is greater than one, and its only positive factors are one and itself. The following is a (sketch of a) well known proof of the fact that there are infinitely many primes.

Proof. Assume that there are only finitely many primes. Then we may list them: $p_1, p_2, \ldots, p_n$. Let $P = (p_1 \cdot p_2 \cdot \cdots \cdot p_n) + 1$. P is greater than all of the p_i so it is not prime. Since it is not prime, it can be written as the product of primes, (this fact is called the fundamental theorem of arithmetic). In particular, it can be written as $P = Qp_i$ for some p_i. But then p_i divides P and $P - 1$ so divides $P - (P - 1) = 1$. This is impossible, because $p_i \geq 2$. Our assumption that there are only finitely many prime must, therefore, be false. So there are infinitely many primes. $\square$

We wanted to prove a statement 'S is true.' but we instead assumed 'S is false' and showed this was inconsistent. This is called a 'proof by contradiction', and is a common proof technique. Similarily to prove an implication 'if a then b', a useful approach is to show that 'if b is not true, then a is not true'. Why is this okay?

Symbolic logic is useful in deciding if two approaches to a proof are equivalent. We look at symbolic logic with a view to proof, techniques, but take it further than this.

Topics Covered:

- Truth value of primitive and compound statement through truth tables
- Tautology, contradiction, logical equivalence, laws of logic, rules of substitution
- Contrapositive, inverse, converse
- Valid argument, premise, conclusion, logical implication, rules of inference
- Quantification, open statement, universe of discourse

2.1 Basic Connectives and Truth Tables

Definition 2.1. A *statement* or *proposition* is a declarative sentence that is either true or false, not both, and it must be decidable which it is.

Examples of statements are '$1 + 1 = 3$', '7 is a quadratic residue modulo 10'. The sentence '$x > 0$' is not a statement, it is neither true or false, but if x is defined as a number, then it becomes a statement.

Statements are usually denoted by lowercase letters. We might let p be the statement $1+1 = 3$, in which case we could determine that p is a false statement.

Definition 2.2. Given a statement p, the *negation* of p, or 'not p', is denoted $\neg p$.

Where p is $1+1 = 3$, $\neq p$ is the statement '$1+1 \neq 3$', which is a true statement.

2.1.1 Basic Connectives

Negation is not the only way we can construct one statement from others. A statement that cannot be broken down into smaller statements is called a *primitive statement* or just a *primitive*.

Given primitives p and q we can construct *compound statements* using any of the following logical connectives. 'And' denoted $\wedge$; 'or' denoted $\vee$; 'xor' or 'exclusive or' denoted $\underline{\vee}$; 'implies' denoted $\rightarrow$; and 'biconditional' or 'if and only if' or 'iff' denoted $\leftrightarrow$.

Statements can be build up from other statements. For example, where p is $1 + 1 = 3$ and q is $1 + 1 = 2$, $p \vee q$ is the statement that '$1 + 1$ is 3 or 2', and is true because at least one of p and q was true.

The truth value, that is, the truth or falsity, of a compound statement depends only on the truth value of the primitives in it. The above connectives are thus rigourously defined by the following table, where we denote truth by 1 and falsity by 0.

p	q	$\neg p$	$p \wedge q$	$p \vee q$	$p \underline{\vee} q$	$p \rightarrow q$	$p \iff q$
0	0	1	0	0	0	1	1
0	1	1	0	1	1	1	0
1	0	0	0	1	1	0	0
1	1	0	1	1	0	1	1

This table is called a *truth table*. They can be used to determine the value of any compound statement from the value of their component primitives. For more complicated statements, we build up the truth table in several steps. The general approach is to approach a statement beginning with 'not' and then bracketed statements, and then working left to right.

Problem 2.1. Make a truth table for the statement $(\neg p \vee q) \to r$.

Solution

p	q	r	$\neg p$	$\neg p \vee q$	$\neg p \vee q \to r$
0	0	0	1	1	1
0	0	1	1	1	1
0	1	0	1	1	1
0	1	1	1	1	1
1	0	0	0	0	0
1	0	1	0	1	1
1	1	0	0	1	1
1	1	1	0	1	1

Observe that in the above example, the statement was true in seven out of eight of the possible truth assignments of the primitives. One thing we are interested in is when a statement is always true, or always false, whatever the truth values of the primitives. From now on, we treat primitives as variable, which can be true or false. Though they are true or false, we don't know what they are.

Definition 2.3. A *tautalogy* is a statement that is always true. A *contradiction* is a statement that is always false.

Example 2.4. The statement $p \vee \neg p$ is a tautology and the statement $p \wedge \neg p$ is a contradiction.

Problem 2.2. Using a truth table, decide whether or not the statement $(p \to (q \to p))$ is a tautology or a contradicion. Can you argue this without a truth table?

One sees that if a statement has n primitives, then the truth table will have 2^n rows. Three primitives is about the most we want to deal with. From here the theory of logical systems that we start to develop will allow us to decide if a statement is a truth or contradiction through more efficient means.

Remark 2.5. Denoting truth by 1 and falsity by 0 exemplifies the analogy between the logic algebra that we are developing, and a boolean algebra. We could further view $\vee$ and $\wedge$ as analogous to addition and multiplication. We won't follow this analogue any more, but if you are familiar with boolean algebras, it might be a useful analogue.

2.2 Logical Equivalence

Problem 2.3. Construct truth tables for the statements '$p \to q$' and '$\neg p \vee q$'.

For every truth assignment of the primitives p and q, these two compound statements have the same truth value.

Definition 2.6. Statements s_1 and s_2 are *logically equivalent* if s_1 is true if and only if s_2 is true. We denote this $s_1 \iff s_2$. If p is a tautalogy, we write $p \iff T_0$ and if it is a contradiction, we write $p \iff F_0$.

Remark 2.7. The notations '$s_1 \leftrightarrow s_2$' and '$s_1 \iff s_2$' mean different things. The first is a logical statement- it may be true or false, and writing it down we are making no assertion about its truth value. The second, '$s_1 \iff s_2$', says that the first, '$s_1 \leftrightarrow s_2$', is a tautalogy.

It is easy to show (though not always easy to determine) that non-equivalent statements are non-equivalent. One only has to provide one line of the truth table- a true assignment of the primitives in which the statements differ.

Example 2.8. Show that $p \wedge (\neg p \to q)$ is not a tautalogy.

Solution

Where $p = 1$ and $q = 0$ we have $\neg p = 1$ so $(\neg p \to q) = 0$ so $p \wedge (\neg p \to q) = 1 \wedge 0 = 0$. Thus the statement is not a tautalogy.

To prove two statements are equivalent, we can always use a truth table, but we would like to avoid doing this too often. So we build up a list of equivalences that we can use over and over.

Above, we saw that

$$p \to q \iff \neg p \vee q$$

so '$\to$' can be expressed in terms of '$\neg$' and '$\vee$'. In the same way, we have

$$\begin{aligned} p \leftrightarrow q \quad &\iff \quad (p \to q) \wedge (q \to p) \\ &\iff \quad (\neg p \vee q) \wedge (\neg q \vee p) \\ &\iff \quad (\neg p \wedge \neg q) \vee (p \wedge q) \end{aligned}$$

and

$$p \underline{\vee} q \iff (p \vee q) \wedge \neg(p \wedge q).$$

Often we will use these equivalences to reduce compound statements to statements using only $\neg$, $\vee$ and $\wedge$.

Problem 2.4. Convince yourself of the logical equivalences given above. Prove that $\wedge$ can also be expressed in terms of '$\neg$' and '$\vee$'.

There are many other useful logical equivalences

Problem 2.5. Use a truth table to show that

$$\neg(p \wedge q) \iff (\neg p \vee \neg q).$$

This is called DeMorgan's law.

We will frequently use the following Laws of Logic.

1	$\neg\neg p \iff p$	Double Negation
2	$\neg(p \vee q) \iff \neg p \wedge \neg q$ $\neg(p \wedge q) \iff \neg p \vee \neg q$	Demorgan
3	$p \vee q \iff q \vee p$ $p \wedge q \iff q \wedge p$	Commutativity
4	$p \vee (q \vee r) \iff (p \vee q) \vee r$ $p \wedge (q \wedge r) \iff (p \wedge q) \wedge r$	Associativity
5	$p \vee (q \wedge r) \iff (p \vee q) \wedge (p \vee r)$ $p \wedge (q \vee r) \iff (p \wedge q) \vee (p \wedge r)$	Distributivity
6	$p \vee p \iff p$ $p \wedge p \iff p$	Idempotence Idempotence
7	$p \vee F_0 \iff p$ $p \wedge T_0 \iff p$	Identity
8	$p \vee \neg p \iff T_0$ $p \wedge \neg p \iff F_0$	Inverse
9	$p \vee T_0 \iff T_0$ $p \wedge F_0 \iff F_0$	Domination
10	$p \vee (p \wedge q) \iff p$ $p \wedge (p \vee q) \iff p$	Absorption
(11)	$p \to q \iff \neg p \vee q$	Implication

Problem 2.6. Read through each of the laws of logic, and convince yourself that they are true. Prove your favourite one using a truth table. Prove that it is your favourite one. Prove that it doesn't pay to choose favourites.

2.2.1 Substitution Rules

On top of the laws of logic, we have some substitution rules which allow us to use them more easily.

- If a statement s is a tautalogy, and every occurence of some primitive p in s is replaced by another statement t, then the resulting statement is a tautology.
- If $s \iff t$, you can always replace s with t in statement without changing the truth values.

Example 2.9. The statement $s : (p \to q) \leftrightarrow (\neg p \vee q)$ is a tautology. Replacing p with $(\neg r \wedge t)$ yields that

$$((\neg r \wedge t) \to q) \leftrightarrow (\neg(\neg r \wedge t) \vee q)$$

is also a tautology.

Example 2.10. By the law of double negation we have:

$$(p \to q) \to \neg p \iff (\neg\neg p \to q) \to \neg p.$$

Here is a more meaty example of the use of the Laws of Logic and the Rules of substitution.

Example 2.11. Show that $(\neg p \to q) \to r \iff (p \to r) \wedge (q \to r)$.

Solution

	$(\neg p \to q) \to r$	
$\iff$	$(\neg\neg p \vee q) \to r$	Substituting $\neg p$ for s and q for t into $s \to t \iff \neg s \vee t$
$\iff$	$(p \vee q) \to r$	Double Negation
$\iff$	$\neg(p \vee q) \vee r$	Subtitution into $s \to t \iff \neg s \vee t$
$\iff$	$(\neg p \wedge \neg q) \vee r$	DeMorgan
$\iff$	$r \vee (\neg p \wedge \neg q)$	Commutativity
$\iff$	$(r \vee \neg p) \wedge (r \vee \neg q)$	Distributivity
$\iff$	$(p \to r) \wedge (q \to r)$	$s \to t \iff \neg s \vee t$

There are a couple more statements that are related to the implication $p \to q$ which get special names as they are used often as proof techniques.

The *contrapositive* of $p \to q$ is $\neg q \to \neg p$. The *converse* of $p \to q$ is $q \to p$. The *inverse* of $p \to q$ is $\neg p \to \neg q$.

Problem 2.7. Show that an implication is logically equivalent to its contrapositive, but not to its converse or inverse. What does this tell you of the relation between the converse and the inverse of an implication?

In proving an implication, one common technique is to prove the contrapositive. When we prove an implication, we often comment on whether or not the converse is true.

Problem 2.8. Where p is the sentence 'It is raining', and q is the sentence 'I skip class', what are the English interpretations of $p \to q$, its contrapositive, its inverse, and its converse.

Let's finish this section with another example of proving the equivalence of statements using our laws of logic.

Example 2.12. Negate and simplify the statement $p \to (\neg q \wedge r)$.

Solution

Leaving the word 'simplify' a little vague, we proceed:

$$\begin{array}{lcl} & \neg(p \to (\neg q \wedge r) & \\ \iff & \neg(\neg p \vee (\neg q \wedge r) & \text{Using } s \to t \iff \neg s \vee t \\ \iff & \neg\neg p \wedge \neg(\neg q \wedge r) & \text{DeMorgan} \\ \iff & p \wedge (\neg\neg q \vee \neg r) & \text{Double Negation then DeMorgan} \\ \iff & p \wedge (r \to q) & \text{Using } s \to t \iff \neg s \vee t \end{array}$$

Observe that we used Double Negation and DeMorgan together in one step, and then in the next step, we used Double Negation without even mentioning it. We allow such shortcuts if it doesn't hinder reading. But don't take them too far.

The text finished this section with some examples of how a logical statement can be modelled by an electronic circuit. We won't look at it, but it you are so inclined, it is an interesting way to visualise the equivalence of statements.

2.3 Inference Rules

2.3.1 Logical Implication

Definition 2.13. An *argument* consists of a conjunction of statements :

$$p_1 \wedge p_2 \wedge \cdots \wedge p_d$$

called the *premises* and a *conclusion* q.

We write an argument as

$$[p_1 \wedge p_2 \wedge \cdots \wedge p_d] \to q.$$

It is called true or *valid* if it is a tautalogy.

Example 2.14. Is the following argument valid?

$$[(p \to r) \wedge (\neg q \to p) \wedge \neg r] \to q$$

Solution

Again, we can use a truth table.

p	q	r	$s_1 : p \to r$	$s_2 : \neg q \to r$	$s_3 : \neg r$	$[s_1 \wedge s_2 \wedge s_3] \to q$
0	0	0	1	0	1	
0	0	1	1	1	0	
0	1	0	1	1	1	1
0	1	1	1	1	0	
1	0	0	0	0	1	
1	0	1	1	1	0	
1	1	0	0	1	1	
1	1	1	1	1	0	

There is only one row in which all of the premises are true, and in this case the conclusion is also true, so the argument is valid.

Example 2.15. Is the argument $[(p \to q) \wedge q] \to p$ valid?

Solution

No! (Don't you dare.) If $q = 1$ and $p = 0$ then $(p \to q) \wedge q$ is 1, but $p = 0$.

Example 2.16. Is the argument $[p \vee \neg p] \to q$ valid?

Indeed it is, as the premises are a contradiction. In this case we say that the argument is *vacuoulsy true*. 'Vacuous' means 'empty', like a vacuum. It is saying, 'sure the argument is true, but this is not really saying anything'.

Definition 2.17. If p and q are statements such that $p \to q$ is a tautology, then we say p *logically implies* q and write $p \Rightarrow q$

Note that $p \to q$ can be true or false, but $p \Rightarrow q$ means $p \to q$ is a tautology. So '$p \iff q$' is the same as '$p \Rightarrow q$ and $q \Rightarrow p$'.

Problem 2.9. Use a truth table to prove that $[(p \to q) \wedge p] \Rightarrow q$.

The logical implication in the above problem is a *inference rule* called *Modus Ponens*. Having proved it, we use it in proving the validity of other arguments.

Example 2.18. To prove the implication $[(p \to q) \wedge \neg q] \Rightarrow \neg p$ we start with the premises, and work to the conclusion.

$$\begin{array}{rcl} & (p \to q) \wedge \neg q & \\ \iff & (\neg q \to \neg p) \wedge \neg q & (contraposative) \\ \Rightarrow & \neg p & (ModusPonens). \end{array}$$

It will make an argument clearer if we write it in a table as follows.

1	$p \to q$	premise (given)
2	$\neg q$	premise
3	$\neg q \to \neg p$	contraposative of 1
$\therefore$	$\neg p$	2 and 3 and Modus Ponens

The symbol $\therefore$ is read 'therefore', and denotes that we have arrived at a conclusion. The implication above is another inference fule called *Modus Tollens*. We have several more inference rules:

	Premises	Conclusion	Name
1	$p \wedge (p \to q)$	q	Modus Ponens
2	$(p \to q) \wedge (q \to r)$	$p \to r$	Syllogism
3	$(p \to q) \wedge \neg q$	$\neg p$	Modus Tollens
4	$(p) \wedge (q)$	$p \wedge q$	Conjunction
5	$(p \vee q) \wedge \neg p$	q	Disjunctive Syllogism
6	$(\neg p \to F_0)$	p	Contradiction
7	$p \wedge q$	p	Conjunctive Simplification
8	p	$p \vee q$	Disjunctive Amplification
9	$(p \wedge q) \wedge [p \to (q \to r)]$	r	Conditional Proof
10	$(p \to r) \wedge (q \to r)$	$(p \vee q) \to r$	Proof by cases
11	$(p \to q) \wedge (r \to s) \wedge (p \vee r)$	$q \vee s$	Constructive Dilemma
12	$(p \to q) \wedge (r \to s) \wedge (\neg q \vee \neg s)$	$(\neg p \vee \neg r)$	Destructive Dilemma

These will help us prove logical implications.

Example 2.19. Show the argument $[(p \to q) \wedge \neg q \wedge \neg r] \to \neg(p \vee r)$ is valid.

Solution

1	$p \to q$	premise
2	$\neg q$	premise
4	$\neg p$	1 and 2 and Modus Tollens
5	$\neg r$	premise
6	$\neg p \wedge \neg r$	conjunction of 4 and 5
$\therefore$	$\neg(p \vee r)$	6 DeMorgan

Problem 2.10. Prove $[(p \to q) \wedge (r \to \neg q) \wedge r] \Rightarrow \neg p$.

Problem 2.11. Prove $[(p \to (q \to r)) \wedge (p \vee s) \wedge (t \to q) \wedge \neg s] \Rightarrow (\neg r \to \neg t)$.

To show that an argument is false, we need only to give a counterexample.

Example 2.20. To see that

1	$p \to q$
2	$\neg p$
$\therefore$	$\neg q$

is an invalid argument, let $p = 0$ and $q = 1$. Then premises 1 and 2 are true, but the conclusion is false.

2.3.2 Common Proof Techniques

Two common proof techniques that we get from our inference rules are 'Proof by Contradiction' and 'Indirect proof.' Proof by Contradiction us a proof that used the rule of contradiction. For this we introduce a false premise, the negation of our conclusion, and show it leads to falsity.

Example 2.21. We show that

1	$(s \wedge h) \to r$
2	$r \to c$
3	$\neg c$
$\therefore$	$\neg s \wedge \neg h$

is valid. Indeed, taking 1 - 3 as premises, we continues as follows

4	$\neg(\neg s \wedge \neg h)$	contradictory premise
5	$s \wedge h$	4 and DeMorgan
6	r	1 and 5 and MP (Modus Ponens)
7	c	2 and 6 and MP
8	$c \wedge \neg c$	3 conjunction 7
9	F_0	8 and Inverse Law
$\therefore$	$\neg s \wedge \neg h$	4 and 9 and contradiction

Since $(p \to (q \to r)) \iff ((p \wedge q) \to r)$, to show

$$[p_1 \wedge p_2 \wedge \cdots \wedge p_n] \to (q \to r)$$

it is enough to show

$$[p_1 \wedge p_2 \wedge \cdots \wedge p_n \wedge q] \to (r).$$

This is called an indirect proof.

Example 2.22. We prove the Rule of Syllogism (see the table) by indirect proof.

1	$p \to q$	premise
2	$q \to r$	premise
3	p	indirect premise
4	q	1,3, MP
$\therefore$	r	2,4 MP

Thus by indirect proof, we get the rule of Syllogism.

Problem 2.12. Prove

1	$p \to (q \to r)$
2	$\neg q \to \neg p$
$\therefore$	$p \to r$

2.4 Quantifiers

Recall that statements such as

$$p(x) : x \text{ is a multiple of } 3,$$

and

$$q(x, y) : x > y,$$

are not **statements**. Because of the variable, their truth values cannot be determined. However, when their variables are replaced with numbers, the become statements.

Definition 2.23. An *open statement* is a sentence that

- contains one or more variable,
- becomes a statement when the variables are replaced with certain allowable choices.

Example 2.24. Where $p(x)$ is the open statement above, $p(6)$ is the true statement '6 is a mutliple of 3', while $p(5)$ is the false statement '5 is a mutliple of 3'.

There is another way to change a open statement into a statement.

Definition 2.25. The allowable choices for the variables of an open statement are called the *universe* or *universe of discourse*, of the open statement. The symbol $\forall$, read as 'for every', is the *universal quantifier*. The symbol $\exists$, read as 'for some', is the *existential quantifiers*. They *quantify* an open statement $r(x)$ with universe U, as follows.

'$\forall x[r(x)]$' is the statement that '$r(x)$ is true for **all** $x \in U$.'

'$\exists x[r(x)]$' is the statement that '$r(x)$ is true for **some** $x \in U$.'

Note

Often, the universe of an open statement does not matter, or can be assumed from 'what is reasonable'. Certainly to interpret $p(6)$ as the statement '6 is a multiple of 3', we don't need to know the universe. But for quantified statements, it is usually important. Indeed, where $p(x)$ is as above $\forall x[p(x)]$ is false if the universe is $U = \mathbb{Z}$, but true if it is $U = 3\mathbb{Z}$.

That said, choosing the universe as $U = 3\mathbb{Z}$ seems a little like trickery. Sometimes even for open statements, if the universe is omitted, we assume it to be the largest reasonable universe. The statement 'x is a multiple of 3' appears very much to be about the integers, it would be reasonable to assume this is the universe.

Example 2.26. Let $q(x, y)$ be the statement $x > y$ have universe $\mathbb{R}$. Then

(a) $\forall x \forall y[q(x, y)]$ is false.

(b) $\forall x \exists y[q(x, y)]$ is true.

(c) $\exists x \forall y[q(x, y)]$ is false.

Hey look! The order of the quantifiers matters.

Problem 2.13. Where the universe is $\mathbb{Z}$, let

- $p(x) : x^2 - 5x + 6 = 0$
- $q(x) : x > 0$
- $r(x) : x^2 - 7 \geq 0$.

Determine the truth value of the following statements.

(a) $\exists x[p(x) \wedge \neg q(x)]$

(b) $\exists x[\neg q(x) \rightarrow p(x)]$

(c) $\forall x[p(x) \rightarrow q(x)]$

(d) $\forall x[p(x) \vee (q(x) \wedge r(x))]$

Problem 2.14. Where the universe of x is the set of students in the class, and the universe of y is a score, let $s(x, y)$ be the open statement:

> x got y on the test.

Hint

You can use arithmetic symbols $<$, $\leq$, and $=$.

Express the following statements symbolically.

(a) Some student got 7.5 on the test.

(b) Every student got at least 5 on the test.

(c) Some student got more than 9 on the test.

(d) No student got 8.5 on the test.

(e) Only one student got 0 on the test.

2.4.1 Quantifications of Logical Equivalences

Recall the logical equivalence

$$p \to q \iff \neg p \vee q.$$

It follows by substitution that if $p(x)$ and $q(x)$ are open statements with the same universe, then

$$p(x) \to q(x) \iff \neg p(x) \vee q(x)$$

for any choice of x in the universe.

Thus the following quantified versions of the logical equivalence are also true:

- $\forall x[p(x) \to q(x)] \iff \forall x[\neg p(x) \vee q(x)]$
- $\exists x[p(x) \to q(x)] \iff \exists x[\neg p(x) \vee q(x)]$

Every logical equivalence has similarily quantified versions.

2.4.2 Interaction of Quantifiers and Connectives

One can show the following.

Proposition 2.27. *(a)* $\forall x[p(x) \wedge q(x)] \iff \forall x[p(x)] \wedge \forall x[q(x)]$

(b) $\exists x[p(x) \wedge q(x)] \Rightarrow \exists x[p(x)] \wedge \exists x[q(x)]$

(c) $\forall x[p(x) \vee q(x)] \Leftarrow \forall x[p(x)] \vee \forall x[q(x)]$

(d) $\exists x[p(x) \vee q(x)] \iff \exists x[p(x)] \vee \exists x[q(x)]$

(e) $\neg\forall x[p(x)] \iff \exists x[\neg p(x)]$

(f) $\neg\exists x[p(x)] \iff \forall x[\neg p(x)]$

Problem 2.15. Convince yourself that the logical implications from Proposition 2.27 are true. Show that $\Leftarrow$ is not true in (b) and $\Rightarrow$ is not true in (c).

The above rules allow us to prove other logical implications involving the quantifiers.

Example 2.28. Show that $\exists x[p(x) \to q(x)] \wedge \forall x[p(x)] \Rightarrow \exists x[q(x)]$.

Solution

$$
\begin{array}{lll}
 & \exists x[p(x) \to q(x)] \land \forall x[p(x)] & \\
\iff & \exists x[\neg p(x) \lor q(x)] \land \forall x[p(x)] & \text{from } s \to t \iff \neg s \lor t \\
\iff & \exists x[\neg p(x)] \lor \exists x[q(x)] \land \forall x[p(x)] & \text{Prop 2.27 (d)}
\end{array}
$$

3 Set Theory

We cover Section 3.1-3.4 of the text, however have rearranged material so our first three sections don't agree exactly with the text. We also include material from Section 5.2 and 5.5 of the text in our Section 3.3.

Topics Covered:

- Set-builder notation, subset, emptyset
- Set operations, set laws, Venn diagrams
- Cardinality, power-set, inclusion-exclusion
- The Pigeonhole principle, counting by bijections
- Probability

3.1 Sets and Subsets

This mostly agrees with Section 3.1 of the text.

3.1.1 Basic Definitions

The notation of symbolic logic allows us to give precise definitions for many notions we have until now defined only intuitively.

We still leave the idea of a set A a bit imprecise: 'a collection of distinct elements from a universe', but given this, we lets look at some related definitions.

Given two sets A and B with the same universe U, A *is a subset of* B, written $A \subseteq B$, if

$$\forall x[x \in A \to x \in B]$$

.

Problem 3.1. Given this definition, is $A \subseteq A$. Where $\emptyset = \{\,\}$ is *the empty set*, is $\emptyset \subseteq A$? Is $\{3, 1, 2, 2\} \subseteq \{1, 2, 3\}$.

How would you define the statement $A = B$, that the sets A and B are the same set? One way would be

$$\forall x[x \in A \iff x \in B]$$

.

But look at the following logical equivalence:

$$\begin{aligned}\forall x[x \in A \iff x \in B] \quad &\iff \quad \forall x[x \in A \to x \in B] \wedge \forall x[x \in B \to x \in A] \\ &\iff \quad A \subseteq B \wedge B \subseteq A\end{aligned}$$

This rigourously proves the obvious statement that

$A = B$ if and only if $A \subseteq B$ and $B \subseteq A$.

As simple as this seems, we will use it a lot in proofs.

A set A is a *proper subset* of B, written $A \subset B$ if $A \subseteq B$ and $A \neq B$.

Give and alternate definition of $A \subset B$:

$A \subseteq B$ and _____.

As the basic set definitions have 'quantified open statement' versions, proofs about set theory will be very similar to proofs of logical equivalences and logical implications. Let's see one.

Theorem 3.1. *Let $A, B, C \subseteq U$.*

i. If $A \subseteq B$ and $B \subseteq C$ then $A \subseteq C$.

ii. If $A \subset B$ and $B \subseteq C$ then $A \subset C$.

iii. If $A \subseteq B$ and $B \subset C$ then $A \subset C$.

iv. If $A \subset B$ and $B \subset C$ then $A \subset C$.

Proof. We prove only *i* and *iii*. To prove *i*, assume the premises, that $A \subseteq B$ and $B \subseteq C$. By definition this means that

$$\begin{aligned} & \forall x[x \in A \to x \in B] \wedge \forall x[x \in B \to x \in C] \\ \iff \quad & \forall x[(x \in A \to x \in B) \wedge (x \in B \to x \in C)] \\ \Rightarrow \quad & \forall x[(x \in A \to x \in C] \text{ by syllogism.}\end{aligned}$$

This last statement is the definition of $A \subseteq B$.

Now we prove *iii*. Assume that $A \subseteq B$ and $B \subset C$. As $B \subset C$ implies $B \subseteq C$ we have the premises of part *iii*., so we may conclude that $A \subseteq C$. Thus we only have to show that there is some x that is in C but not A. By the premise $B \subset C$ there is some x in C that is not in B. As A is a subset of B, this x is not in A either. So we are done. $\square$

Notice that our proof style differed from part *i* to part *c*. In *a* we were very notational while in *c* we relied a lot more on english sentences. Which one is easier for you to read? Mathematics tends to be written as in part *c*, as it is easier to read. But it is still precise, many words or phrases have more restricted, or different usage in Math than they do in regular conversation. It is good to learn some of these usages.

Note on English

The word 'assume' is usually comes at the beginning of a proof and will be followed, often much later, by

- a conclusion, if we are proving an argument to be valid; or
- a contradiction, when we are doing proof by contradiction.

When the proofs involved between the assumptions and the conclusion/contradiction are very short, we often use 'if' instead of 'assume'. Ex. 'If x is in A then it is also in B, which is false, so x is not in A.'

Problem 3.2. Finish the proof of Theorem 3.1. Try to write it more like the proof of part *iii* above.

3.1.2 Set-builder Notation

As for these definitions, universal quantification is usefel in describing specific sets. This is seen With *set-builder* notation. Given a universe U and an open statement $p(x)$,

$$\{x \in U \mid p(x)\}$$

(or simply $\{x \mid p(x)\}$ when the universe is understood), is the set of elements in U for which p is true.

For example the set $2\mathbb{N} = \{0, 2, 4, 6, \dots\}$ can be described by $2\mathbb{N} = \{x \in \mathbb{Z} \mid x \geq 0, 2 \text{ divides } x\}$.

As a variation that works when the universe is understood, one can even use a function of x: $2\mathbb{N} = \{2x \mid x \geq 0\}$.

3.2 Set Operations and Laws of Set Theory

Mostly Section 3.2 and 3.3 of the text.

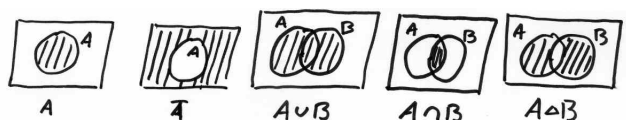
Each of the basic logical connectives defines an operation on sets.

Let A and B be sets with the same universe U, then we have

- the *complement* $\overline{A} = \{x \mid x \notin A\}$
- the *union* $A \cup B = \{x \mid x \in A \vee x \in B\}$

- the *intersection* $A \cap B = \{x \mid x \in A \wedge x \in B\}$, and
- the *symmetric difference* $A \triangle B = \{x \mid x \in A \underline{\vee} x \in B\}$.

These set operations can be visualised by a *Venn (or circle) Diagram.*



There is also the *relative complement of A in B* defined as

$$A - B = \{x \mid x \in A \vee x \notin B\}.$$

Problem 3.3. Prove that $A - B = A \cap \overline{B}$

Problem 3.4. What is the set $\{x \mid x \in A \to x \in B\}$? Perhaps use the implication law. Draw a Venn diagram to describe it. How does this relate to $A - B$.

As the sets operations are defined from logical connectives, it makes sense that they exhibit similar properties, such as DeMorgan's Law:

$$\overline{A \cup B} = \overline{A} \cap \overline{B}$$

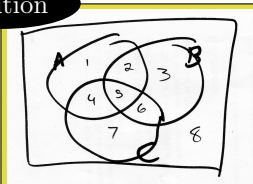
The proof of this follows directly from DeMorgans Law for logic:

$$\begin{aligned}
\overline{A \cup B} &= \overline{\{x \mid x \in A \vee x \in B\}} && \text{def of union} \\
&= \{x \mid \neg(x \in A \vee x \in B)\} && \text{def of complement} \\
&= \{x \mid x \notin A \wedge x \notin B\} && \text{DeMorgan} \\
&= \{x \mid x \in \overline{A} \wedge x \in \overline{B}\} && \text{def of complement} \\
&= \overline{A} \cap \overline{B}
\end{aligned}$$

Similarily, we get the Set Laws, set theoretic versions of all of the laws of logic. See the table on page 139 of the text. It is useful to also include the identity from Problem 3.3, which we view as an analogue of the implication law, in this list.

Example 3.2. Show that $A - (B \cup C) = (A - B) \cap (A - C)$.

Solution



One way to do this is to use a Venn diagram. One can check that both sides of the equation are region 1 in the figure on the left. Alternately we can use the set laws.

$$\begin{aligned} A-(B\cup C) &= A\cap\overline{B\cup C} && \text{Implication Law}\\ &= A\cap\overline{B}\cap\overline{C} && \text{DeMorgan}\\ &= A\cap A\cap\overline{B}\cap\overline{C} && \text{Idempotence}\\ &= (A\cap\overline{B})\cap(A\cap\overline{C}) && \text{Commutativity}\\ &= (A-B)\cap(A-C) && \text{Implication} \end{aligned}$$

3.3 More on counting

This pulls from Sections 3.1, 5.2, 5.5

3.3.1 Double-Counting

The *cardinality* $|A|$ of a finite set A is the number of elements in the set. For infinite sets such as $\mathbb{Z}$ or $\mathbb{R}$, we say the cardinality is ∞. In other classes you will see that there are various infinite cardinalities, but that won't interest us in this class.

The power set of $\mathscr{P}(A)$ of a set A is the set of all subsets of A. If $|A| = n$, what is $|\mathscr{P}(A)|$?

One easy way to count it is using the product rule. In choosing an element of $\mathscr{P}(A)$, that is, a subset S of A, we have to choose for each of the n elements x of A whether or not x is in S. There are 2^n ways to do this, so $\mathscr{P}(A) = 2^n$.

Another way to count is to use the sum rule. Counting the 0 element sets, then the 1 element sets, et cetera, we get $|\mathscr{P}(A)| = \binom{n}{0} + \binom{n}{1} + \cdots + \binom{n}{n}$. Thus

$$2^n = \sum_{i=0}^{n} \binom{n}{i}.$$

The proof we have given of this identity is an example of what is called *double counting*. We counted a set in two different ways to arrive at the identity. This is a common technique for proving combinatorial indentities.

Problem 3.5. Show that this identity also follows from the Binomial theorem.

Problem 3.6. Show that

$$\binom{n+1}{r} = \binom{n}{r} + \binom{n}{r-1}$$

by double counting the number of r-combinations of an $n+1$ element set.

3.3.2 Counting by Bijections

We are good at counting permutations and combinations of various types. It makes sense to relate other counting problems to these problems. We saw this when counting the number of positive integer solutions to the equation $x_1 + x_2 + x_3 = 10$, but it is a fundamental idea.

Given two sets A and B, if we can define a *bijection* (a 1-to-1, onto function) from A to B. Then $|A| = |B|$.

Even when we know the cardinality of a set B it can be useful to get a bijection with another well know set A.

Example 3.3. We define a bijection f from the powerset $\mathscr{P}([n]$ to the set of n digit binary strings by letting the ith digit of $S \in \mathscr{P}([n])$ be 1 if $i \in S$ and 0 otherwise.

We can verify that it is a bijection by exhibiting an *inverse* function g from the set of n-digit binary strings to $\mathscr{P}([n])$: $g(b_n b_{n-1} \ldots b_2 b_1)$ contains the element i if and only if b_i is 1.

Why is this useful, now we have an *ordering* of $\mathscr{P}([n])$ by converting these strings to the integers they represent. If we want to write an algorithm checking some property of subsets of $[n]$, this is useful.

A *composition* of the number n is an expression of n as a sum of positive integers. So $5 = 1+1+1+1+1$, $5 = 1+2+1+1$, $5 = 2+1+1+1$, $5 = 4+1$. How many compositions are there of the number n?

We could count the number of compositons with i summands for each i, and then add them up. (See Section 1.4 of the text.) But we will count the set by bijections.

Indeed, we will show that there are 2^{n-1} compositions of n by giving a bijection between the set of compositions, and $\mathscr{P}([n-1]$.

Start with the string $(1+1+\cdots+1+1)$. There are $n-1$ occurences of the symbol '+'. For a set $S \in \mathscr{P}([n-1])$, let $f(S)$ be the composition constructed as follows. If i is in S replace the ith occurence of '+' with ')+('. So the subset $1, 3, 4 \subset [8]$ corresponds to the string

$$(1) + (1+1) + (1) + (1+1+1+1).$$

Then collapse the bracketed sums: $1+2+1+4$.

One can check that this is a bijection.

Problem 3.7. Using the ordering of $\mathscr{P}([7])$ you get from the bijection of it to the 7-digit binary strings, give the first three compositions of [8].

3.3.3 k - to - 1 functions

Generalising the idea of counting by bijections or 1-to-1 functions, we can count by k-to-1 functions.

Given two sets A and B, if we can define a function $f : A \to B$ such that for each b in B, there are k elements a of A with $f(a) = b$, then $|A| = k|B|$.

We saw this in counting the number of combinations of a set, though we didn't explicitly talk about a function.

To count the number of k-combinations of n we define a function

$$f : (x_1, x_2, \ldots, x_k) \mapsto \{x_1, x_2, \ldots, x_k\}$$

from the set of k-permutations to the set of k-combinations. For each k-combination C, there are $k!$ different k-permutations σ with $f(\sigma) = C$. So f is $k!$-to-1.

We concluded that the number $C(n, k)$ of k-combinations was

$$C(n, k) = \frac{P(n, k)}{k!} = \frac{n!}{k! \cdots (n-k)!}.$$

3.3.4 The Pigeonhole Principle

A variation on the idea of counting by bijections it the idea that if we can define an onto function f from A to B, that is not a bijection, then $|A| > |B|$.

It is usually stated in the converse, with the following example:

> If there are $n + 1$ pigeons in n pigeonholes, then some hole contains more than one pigeon.

This is a fairly obvious example, but the pigeonhole priciple can be used in more complicated problems.

Problem 3.8. Show that for any 5 points in the equilateral triangle of side length 1, there must be two whose distance apart is at most 1/2.

Example 3.4. A golfer plays 132 games over 11 weeks, playing at least one game a day. Show that there is some stretch of consecutive days in which she plays exactly 21 games.

Solution

Let a_i be the number of games she plays by the end of the ith day. The 154 numbers $a_1, a_2, \ldots, a_{121}$, and $b_1 = a_1 + 21, b_2 = a_2 + 21, \ldots, b_{121} = a_{121} + 21$ are all between 1 and 153, so some two of them are the same. By the condition that she plays at least 1 game a day, all of the a_i are distinct, and all of the b_i are distinct, so there are some i and j such that $b_i = a_j$. So the golfer plays 21 games from the $i + 1^{th}$ to the jth day.

Problem 3.9. At a party, any two people are either acquainted, or are strangers. How many people must there be at the party so that there are either some three people that are mutually acquainted, or there are some three people that are mutual strangers.

3.3.5 Inclusion and Exclusion

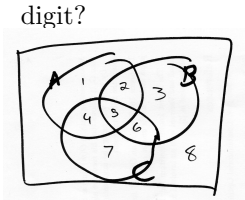
Observe from the Venn diagram that $|A \cup B| = |A| + |B| - |A \cap B|$. We can use this to count the number of integers in [60] that are multiples of 2 or 3:

Solution

Let A be the set of multiples of 2 and B be the set of multiples of 3. Then $|A| = 30$ and $|B| = 20$ and $A \cap B$ is the set of multiples of 6, so $|A \cap B| = 10$. So by the above identity we have that

$$|A \cup B| = |A| + |B| - |A \cap B| = 30 + 20 - 10 = 40$$

Problem 3.10. How many arrangements of the [6] start or end with an odd digit?



What about for more circles in the Venn Diagram? We see that $|A \cup B \cup C|$ is equal to

$$\begin{aligned} & |A| + |B| + |C| \\ - \quad & |A \cap B| - |A \cap C| - |B \cap C| \\ + \quad & |A \cap B \cap C| \end{aligned}$$

Example 3.5. Let's count the number of arrangements of the letters of the word CHEMIST that have H before T, or T before E, or E before M.

Let A_1 be the set of arrangements with H before T. Let A_2 be the set of arrangements with T before E. Let A_3 be the set of arrangements with E before M.

Then $|A_1| = \frac{7!}{2!} = |A_2| = |A_3|$, $|A_1 \cap A_2| = \frac{7!}{3!} = |A_2 \cap A_3|$, $|A_1 \cap A_3| = \frac{7!}{2!2!}$, and $|A_1 \cap A_2 \cap A_3| = \frac{7!}{4!}$.

Using the above identity, we can put this together to get $|A_1 \cap A_2 \cap A_3|$.

3.4 A First Word on Probability

Probability is a convenient way of counting the relative sizes of sets.

Definition 3.6. An *experiment* is an activity with some set of possible outcomes. The *sample space* is the set S of possible outcomes. An *event* is a subset A of the sample space.

Example 3.7. In an experiment we toss two coins. Each coin can come up 'Heads' (H) or 'Tails' (T), so our sample space is $S = \{HH, HT, TH, TT\}$. Some possibles events are

- $A_1 = \emptyset$.
- $A_2 = \{HT, TH\}$.
- $A_3 = S$.

If an experiment has a sample space in which each outcome is equally likely, the probability $Pr(A)$ of an event A is

$$Pr(A) = |A|/|S|.$$

We write $Pr(a)$ for $Pr(\{a\})$.

Example 3.8. Continuing the previous example, we have that $Pr(A_1) = 0$, $Pr(A_2) = 1/2$, and $Pr(A_3) = 1$.

Example 3.9. Two distinguishable dice are rolled. Find the probability that the rolled numbers sum to a) 7, b) 9, c) 12. d) Find the probability that at least one 2 is rolled.

Solution

The sample space S has 36 elements.
a) $Pr(\{16, 25, 34, 43, 52, 61\}) = \frac{6}{36} = 1/6$
b) $Pr(\{36, 45, 54, 63\}) = \frac{4}{36} = 1/9$
c) $Pr(\{66\}) = 1/36$
d) Let A_1 be the event that the first die is a 2 and let A_2 be the event that the second die is a 2. The we are looking for

$$\begin{aligned} Pr(A_1 \cup A_2) &= \frac{1}{36}|A_1 \cup A_2| \\ &= \frac{1}{36}(|A_1| + |A_2| - |A_1 \cap A_2|) \\ &= \frac{1}{36}(6 + 6 - 1) = \frac{11}{36}. \end{aligned}$$

Problem 3.11. You choose 2 Jellybellies from a cup of 30 different Jellybellies. What is the probability of choosing

a) banana and lemon

b) neither banana or lemon

c) banana but not lemon

Problem 3.12. In a group of 80 students

- 45 take Math
- 48 take CompSci
- 38 take Stats
- 30 take Math and CompSci
- 30 take CompSci and Stats
- 23 take Math and Stats
- 15 take all three

a) How many students take none of Math CompSci or Stats?

b) Find the probability that a randomly chosen student takes Math.

c) Find the probability that a randomly chosen Stats student also takes CompSci.

4 Induction and the Integers

In this chapter we cover all sections of Chapter 4 of the text. We also add some material about the 'modulo' function.

Topics Covered:

- Inductive proofs
- Recursive definitions
- The Fibonacci numbers
- Divisibility and the Division Algorithm
- Prime numbers
- The (Reverse) Euclidean Algorithm and GCD
- The Fundamental Theorem of Arithmetic
- The integers modulo m.

4.1 Induction

A *proof by induction* proceeds as follows. For a open statement $S(n)$ where the universe for n is the set of positive integers, we can prove that $S(n)$ is true for all n by proving the following two statements.

i. $S(1)$ is true.

ii. For all $k \geq 2$, if $S(k-1)$ is true, then $S(k)$ is true.

Having done this, the principle of induction impies that $S(n)$ is true for all n.

Statement 1. is called the 'basis step' and 2. is called the 'induction step'. The assumption that $S(k-1)$ is true, in the induction step, is called the 'induction hypothesis'.

See Examples 4.1 and 4.4 of the text.

In practice, the universe for n may vary. But as long as there is a nice bijection between the universe of n and $\mathbb{Z}^+$, we can still do induction.

See Example 4.10 of the text, for example, where the universe of n is $\mathbb{N}$ instead of $\mathbb{Z}^+$. The principle of induction still applies here, indeed, we could replace every occurence of n with $n-1$ and 'shift' our argument to the universe $\mathbb{Z}^+$.

4.1.1 Strong induction

Sometime it is useful strenghen our induction assumption, and assume that a statement $S(n)$ has been proved for more values of n.

See Example 4.13 of the text

4.2 Recursive Definitions

We can define things using a process similar to induction. This process is called recursion, it consists of a base step, and a recursion step.

The following three definitions are by recursion.

Definition 4.1. Let the notation $n!$ be defined as follows.

- Base Step: $0! = 1$
- Recursion Step: For $n \geq 1$, $n! = (n-1)! \cdot n$.

Definition 4.2. Let the general union $A_1 \cup A_2 \cup \cdots \cup A_n$ be defined for all $n \geq 2$ as follows.

- $A_1 \cup A_2 = \{x \mid x \in A_1 \vee x \in A_2\}$
- $A_1 \cup A_2 \cup \cdots \cup A_n = (A_1 \cup A_2 \cup \cdots \cup A_{n-1}) \cup A_n$

In any course in which recursion is defined, one must have the following example.

Definition 4.3. The nth Fibonacci number F_n is defined for all $n \geq 0$ as follows.

- $F_0 = 1$ and $F_1 = 1$.
- For $i \geq 2$, $F_n = F_{n-2} + F_{n-1}$.

Recursive definitions lend to inductive proofs.

See Example 4.18 of the text, proving a general version of DeMorgan's Law

Recall the identity about binomial coefficients that we've proved a couple of times.

$$\binom{n+1}{r} = \binom{n}{r} + \binom{n}{r-1}.$$

We can use this to recursively define the binomial coefficients as follows.

Definition 4.4. The binomial coefficients are defined by

- $\binom{0}{0} = 1$
- $\binom{n}{r} = 0$ if $r > n$ or $r < 0$
- $\binom{n+1}{r} = \binom{n}{r} + \binom{n}{r-1}$ otherwise

4.3 The Division Algorithm

Definition 4.5. For integers a and b, we say 'a *divides* b', and write $a \mid b$, if there is some integer x such that $ax = b$. In this case, we also say a is a *divisor* or *factor* of b, and b is a *multiple* of a.

Example 4.6. Accoring to this definition we have that: $3 \mid 9$, $-2 \mid 12$, $2 \mid -12$, $7 \mid 0$, '$0 \mid a \Rightarrow a = 0$'.

Further we have $a \mid 0 \Rightarrow a = 0$. This is an important property of the integers. It is the property of having 'no zero divisors', and doesn't hold in such number systems as $\mathbb{Z}_{12}$: the integers modulo 12. This is introduced in Chapter 14 of the text. We will not look at $\mathbb{Z}_m$, but in the end of this chapter look at something very similar.

Anyways, without this property, we would not be able to cancel:

$$ab = ac \Rightarrow b = c$$

and many of the properties we prove in this chapter would not be true.

Let's record/prove a couple easy properties dealing with divisiblity.

Theorem 4.7. *For all integers a, b, and c the following hold.*

a) $1 \mid a$, $-1 \mid a$, *and* $a \mid 0$.

b) If $a \mid b$ and $b \mid a$, then $a = \pm b$.

c) If $a \mid b$ and $b \mid c$, then $a \mid c$.

d) If $a \mid b$ then $a \mid xb$ for all integers x.

e) If $x + y = z$ and a divides two of x, y and z, then a divides all of x, y, and z.

f) If a divides c_i for $i = 1, \ldots, d$, then a divides any linear combination *of the c_i, that is:*

$$a \mid x_1c_1 + x_2c_2 + \cdots + x_dc_d$$

for any integers $x_1, \ldots, x_d$.

Proof. You should be able to prove all of these, but we prove only b) and e).

To see b), assume that $a \mid b$ and $b \mid a$. By definition, there are integers x and x' such that $ax = b$ and $bx' = a$, and so $a = bx' = (ax)x' = a(xx')$. This implies that $xx' = 1$, but both x and x' are integers so $x = x' = \pm 1$. From $a = x'b$ this give $a = \pm b$.

For e), there are three cases to prove, for the first, we assume that a divides x and z and show that it divides y also. The proofs of the other cases (what are they?) are essentially the same, so we omit them. Assuming that a divides

x and z, we have that $ax' = x$ and $az' = z$ for some integers x' and z'. Now $y = z - x = az' - ax' = a(z' - x') = ay'$ where $y' = (z' - x')$ is an integer. So a divides y. $\square$

The following two problems use the above rules.

Problem 4.1. Do there exist integers x, y and z such that

$$7x + 14y + 49z = 100?$$

Solution

No. We know that 7 divides all of $7, 14$, and 49, so it divides the left side of the equation, but it doesn't divide the right side: 100.

Problem 4.2. Suppose that $13 \mid 3a + 4b$. Show that $13 \mid 7a + 5b$.

Solution

From Theorem 4.7 d) we have that

$$13 \mid -2(3a + 4b) = -6a - 8b.$$

From the fact that $13 \mid 13$, we have using Theorem 4.7 that

$$13 \mid 13a + 13b.$$

Thus by Theorem 4.7 b),

$$13 \mid 13a + 13b - 6a - 8b = 7a + 5b,$$

as needed.

Definition 4.8. An integer $p \geq 2$ is *prime* if its only positive divisors are 1 and p. It is *composite* if it isn't prime.

Note that 1 is neither prime or composite. Also note that by definition any composite number n can be expressed as the product of two integers greater than 1. In fact with a bit of work we can say that a can assumed to be prime.

Lemma 4.9. *Let $n \geq 2$ be an integer. Then there is some prime p such that $p \mid n$.*

Proof. Our proof is by strong induction. For $n = 2$ we have $2 \mid 2$, and 2 is prime, so we are done. Assume the lemma has been proved for all values less then n. If n is prime, then $n \mid n$ and we are done, so we may assume that n is composite. Then $n = ab$ for some integers a and b less than n. By the induction hypothesis, $p \mid a$ for some prime p. But then $p \mid ab = n$ by Theorem 4.7. $\square$

This lemma helps us prove the following old theorem of Euclid.

Theorem 4.10. *There are infinitely many primes.*

Proof. Towards condtradiction, assume that there are finitely many primes: $p_1, \ldots, p_n$. Let $N = p_1 p_2 \ldots p_n + 1$. By Lemma 4.9 there is some prime q such that $q \mid N$. Since it is prime, $q = p_i$ for some i, so $q \mid p_1 p_2 \ldots p_n$. By Theorem 4.7 we have that $q \mid 1$. But this implies that $q = 1$, a contradiction. $\square$

Theorem 4.11 (The Division Algorithm)**.** *For $a, b \in \mathbb{Z}$ with $b > 0$, there exist unique $q, r \in \mathbb{Z}$ with $0 \leq r < b$ such that $a = qb + r$.*

Proof. We split the proof up into two parts: the existence of q, r, and their uniqueness.

For existence, we first prove the theorem for $a \geq 0$ by induction on a. For $0 \leq a < r$ and any b, $q = 0$ and $r = a$ make the equation true. This is our base case. Now fix $b \geq 0$ and let $a \geq r$. Assume that for any non-negative integer a' less then a we have $a = bq' + r'$ for integers q' and r' with $0 \leq r' < b$. Then because $a > r$, we have that $a - r = q'b + r'$, and so $a = (q' + 1)b + r'$.

Now for $a < 0$, $-a = q'b + r'$ for some q' and $0 \leq r' < b$, and so $a = -q'b - r'$. If $r' = 0$ we are done, otherwise, $a = (-q' - 1)b + b - r' = (-q' - 1)b + r$ where $r = b - r'$ is in $\{1, 2, \ldots, b - 1\}$.

To see that q and r are unique, for given a and b, we assume that $qb + r = a = q'b + r'$ are two such representations, and show that $q = q'$ and $r = r'$. Indeed, assume wlog that $q \geq q'$. Rearranging $qb + r = q'b + r'$ we get $r' - r = b(q - q')$. As $r' < b$ and r is positive, $b > r' - r$. So $q - q' < 1$. But q and q' are integers with $q \geq q'$ so $q = q'$. By $r' - r = b(q - q')$, this implies that $r = r'$, as needed. $\square$

Although the above theorem is not an algorithm, it is usually called 'The Division Algorithm'. The inductive proof of existence suggests an algorithm for finding q and r when a is positive: subtract b repeatedly from a until you are left with $r < b$. The number of times you did this is q. Our proof ensures that this process terminates, and that the result is unique.

4.3.1 Integers in other bases

Definition 4.12. For integer $b \geq 2$ and $d_0, \ldots d_n \in \{0, 1, \ldots b - 1\}$, the *base b integer* $(d_n d_{n-1} \ldots d_1 d_0)_b$ represents the the integer

$$d_n b^n + d_{n-1} b^{n-1} + \ldots d_2 b^2 + d_1 b + d_0.$$

So, for example, the number 99, which is represented in base 10, can be represented in base 7 as $(201)_7$, or in base 2 as $(1100011)_2$. In base 16, we need extra

digits. It is standard that $(A)_{16}, (B)_{16}, \ldots, (F)_{16}$ are used for $10, 11, \ldots, 15$ respectively. Thus $90 = (5A)_{16}$.

The division algortihm ensures that any integer has a unique base b representation for any $b \geq 2$. Further, it suggests how to find it.

Example 4.13. To represent 131 in base 7 we repeatedly apply the division algorithm by dividing 7 into 131:

$$\begin{aligned} 131 = 18(7) &\quad + \quad 5 \\ 18 = 2(7) &\quad + \quad 4 \\ 4 = 0(7) &\quad + \quad 2 \end{aligned}$$

Reading up the column on the right, we get that $(245)_7 = 131$. Indeed $131 = 7(18) + 5 = 7\,(7(2) + 4) + 5$, which gives that $131 = 2 \cdot 7^2 + 4 \cdot 7 + 5 = (245)_7$.

4.4 GCD and the Euclidean Algorithm

You have probably seen that the 'greatest common divisor', or the 'gcd' of two numbers is the greatest integer c that divides them both. However, our formal definition will use a more useful condition in place of 'greatest'.

Definition 4.14. For two non-zero integers a and b, $\gcd(a, b)$ is the unique positive integer c satisfying all of the following conditions.

i) $c|a$

ii) $c|b$

iii) If $d|a$ and $d|b$, then $d|c$.

It is clear that there are integers c satisfying items i) and ii) indeed 1 does, but not so clear that there is one that satisfies iii) as well. I think we believe it but we haven't talked about the prime decomposition of an integer yet, so it isn't fair to use this.

Theorem 4.15. *For non-zero integers a and b, $\gcd(a, b)$ exists.*

Proof. Consider the set S of positive integers that can be written as $xa + yb$ for integers x and y. Let $c = x_0a + y_0b$ be the smallest integer in S. We claim $c = \gcd(a, b)$. First we show that $c \mid a$. Assume $c \nmid a$. Then $a = qc + r$ for integers q and r where $r < c$. So $r = a - qc = a - q(x_0a + y_0b) = (1 - qx_0)a + (-qy)b$ is in S. This contradicts the fact that c was the smallest element of S. So $c \mid a$. The proof that $c \mid b$ is the same. Now proving item iii) is easy. If $d \mid a$ and $d \mid b$ then $d \mid xa + yb$ for any integers x and y. In particular, $d \mid x_0a + y_0b = c$. $\square$

Now to find the GCD of two numbers, we can list out the divisors of each number, and scan the lists:

Example 4.16. Finding $\gcd(18, 30)$.
Positive divisors of 18: $1, 2, 3, 6, 9, 18$
Positive divisors or 30: $1, 2, 3, 5, 6, 10, 15, 30$
By inspection, we have $\gcd(18, 30) = 6$.

However, this gets to be a lot of work for large numbers. We look at an algorithm to do it quickly, based on the following lemma.

Lemma 4.17. *If* $a = bq + r$ *then* $\gcd(a, b) = \gcd(b, r)$.

Proof. Let $a = bq + r$. We show that $\gcd(a, b) \mid \gcd(b, r)$ and that $\gcd(b, r) \mid \gcd(a, b)$. The lemma follows because $(x \mid y) \wedge (y \mid x) \Rightarrow x = \pm y$, and gcd are always positive.

By definition, $\gcd(a, b)$ divides a and b. So it also divides $r = a - bq$. As it divides b and r, the definition of $\gcd(b, r)$ implies that $\gcd(a, b) \mid \gcd(b, r)$. The proof that $\gcd(b, r) \mid \gcd(a, b)$ is similar. $\square$

With this lemma in hand, we have a nice algorithm for finding the gcd of two integers. It is called the Euclidean Algorithm, and we introduce it by example.

Example 4.18. The Euclidean Algorithm to find $\gcd(132, 55)$.

Observe that

$$\begin{aligned} 132 = 2(55) &+ 22 \\ 55 = 2(22) &+ 11 \\ 22 = 2(11) &+ 0 \end{aligned}$$

Thus by the lemma, $\gcd(132, 55) = \gcd(55, 22) = \gcd(22, 11) = \gcd(11, 0) = 11$.

In fact, one can take the algorithm further and find the integers x and y such that $\gcd(a, b) = xa + yb$. We proved such integers existed in the proof of Theorem 4.15.

Example 4.19. The Reverse Euclidean Algorithm.

From the calculations of the last example, we have that

$$\begin{aligned} 11 &= 55 - 2(22) \\ &= 55 - 2(132 - 2(55)). \end{aligned}$$

Collecting terms on the right, we have $11 = (-2)132 + (5)55$.

Problem 4.3. Find $\gcd(288, 89)$ and write it as a linear combination of 288 and 89.

By far the most important application of this algorithm is in defusing bombs, as we saw when Bruce Willis and Samuel L. Jackson's characters had to defuse a bomb in Die Hard: With A Vengeance. Essentially, their problem was this:

Problem 4.4. You have 2 containers, one measures 132ml and the other 55ml. How do you use these containers to measure out 11ml? 22ml? What quantities can you measure with these two containers? (See Example 4.37 of text.)

Definition 4.20. Integers a and b are *relatively prime* if $\gcd(a, b) = 1$.

Equivalently, a and b are relatively prime if and only if there exists integers x and y such that $ax = by = 1$. To determine if two integers are relatively we can use the Euclidean Algorithm.

Example 4.21. Prove that for all integers n, $13n + 5$ and $5n + 2$ are relatively prime.

Solution

$$\begin{aligned} 13n+5 &= 2(5n+2) &&+(3n+1) \\ 5n+2 &= (3n+1) &&+(2n+1) \\ 3n+1 &= (2n+1) &&+n \\ 2n+1 &= 2(n) &&+1 \end{aligned}$$

So $\gcd(13n + 5, 5n + 2) = 1$.

Recall that a real number r is *rational* if if can be expressed as a fraction of integers: $r = \frac{a}{b}$. Otherwise it is *irrational*.

Problem 4.5. For an integer n, show that if $\sqrt{n}$ rational then n is a perfect square.

Solution

Assume that $\sqrt{n} = a/b$ for some integers a and b. We may assume that $\gcd(a, b) = 1$, because if it isn't, we can replace a and b by $a/\gcd(a, b)$ and $b/\gcd(a, b)$ respectively. Now $n = a^2/b^2$, so $nb^2 = a^2$.

We show that $b = 1$ and so $n = a^2$, a perfect square. Towards contradiction, assume that $p \mid b$ for some prime p. Then $p^2 \mid b^2 n = a^2$, so $p \mid a$. This contradicts the fact that $\gcd(a, b) = 1$.

As a consequence, we have that, for example, $\sqrt{2}$ is irrational.

4.4.1 Least Common Multiple

Dual to the GCD is the LCM, or least common multiple.

Definition 4.22. The least common multiple $\text{lcm}(a, b)$ of integers a and b, is the is the unique positive integer c satisfying all of the following conditions.

i) $a \mid c$

ii) $b \mid c$

iii) If $a \mid d$ and $b \mid d$, then $c \mid d$.

Problem 4.6. Mimic the proof of Theorem 4.15 to show that the lcm of two integers does indeed exists. (See Thm 4.9 of text.)

Theorem 4.23.

$$\text{lcm}(a, b) \cdot \gcd(a, b) = ab$$

Proof. We'll prove this in the next section.

□

4.5 The Fundamental Theorem of Arithmetic

Theorem 4.24 (The Fundamental Theorem of Arithmetic)**.** *Every integer $n \geq 2$ can be written uniquely (up to order of the factors) as a product of primes.*

This unique representation of a number of a product of primes is called its *prime factorisation.* For example, the prime factorisation of 124 is

$$124 = 2 \cdot 2 \cdot 31.$$

The proof of the Theorem 4.24 consists of two parts. The first parts is an exercise:

Problem 4.7. Using Lemma 4.9 and induction, show that every integer has a representation as a product of primes.

We finish the proof of Theorem 4.24 by showing uniqueness. We start with a lemma.

Lemma 4.25. *Let a and b be positive integers, and p a prime. If $p \mid ab$, then either $p \mid a$ or $p \mid b$,*

We prove this below, but lets first look at our approach. Let r, s and t be the statements '$p \mid ab$', '$p \mid a$' and '$p \mid b$'. The theorem is the sentence $r \to s \vee t$. What we actully show is $(r \wedge \neg s) \to t$. Prove to yourself that these are logically equivalent.

Proof. If $p \mid a$ then we are done, so we assume that $p \nmid a$ and show that $p \mid b$.

As $p \nmid a$, a and p are relatively prime, so there exist x and y such that $xp+ya = 1$. Multiplying by b we get $p(xb) + y(ab) = b$. But p divides $p(xb)$, and it divides $y(ab)$ (because it divides ab), so p divides b, as needed.

□

Using induction on the previous lemma, we can show

Corollary 4.26. *For all positive integers $a_1, \ldots, a_n$, and all prime p, if*

$$p \mid a_1 a_2 \ldots a_n,$$

then $p \mid a_i$ for some $i = 1, \ldots, n$.

Now we prove the uniqueness part of Theorem 4.24.

Proof of uniqueness in Thm. 4.24. The proof is trivially true if $n = 2$. Assume it is true for $n < k$, we show it is true for $n = k$.

Assume that k has two representations as a product of primes:

$$p_1 p_2 \ldots p_m = k = q_1 q_2 \ldots q_n.$$

We want to show that as multisets $\{p_1, p_2, \ldots, p_m\} = \{q_1, \ldots q_n\}$.

Now $p_1 \mid k = q_1 q_2 \ldots q_n$, so by Corollary 4.26, $p_1 | q_i$ for some i in $1, \ldots, n$. As they are both prime, $p_1 = q_i$. By the induction hypothesis, the theorem is true for k/p_1, and so the multisets $\{p_2, p_3, \ldots, p_m\}$ and $\{q_1 q_2 \ldots q_{i-1} q_{i+1} \ldots q_n\}$ are equal. Adding $p_1 = q_i$ to each, they are still the same. □

In the prime factorisation of an integer such as $124 = 2 \cdot 2 \cdot 31$ where a prime occurs more than once, we usually write the factorisation as

$$124 = 2^2 \cdot 31$$

and say the prime 2 has *power* 2 in the factorisation of 124.

If $p^d \mid a$, for some prime p, then $a = p^d \cdot x$ for some integer x and $a = p^d(p_1^{d_1} p_2^{d_2} \ldots p_m^{d_m})$ where the bracketed part is the prime factorisation of x. This is a prime factorisation of a, and so the power of p in the factorisation of a is at least d.

With this we can prove the following:

Lemma 4.27. *Let a and b be integers. For some prime p, let d_a be the power of p in the prime factorisation of a and d_b be the power of p in the prime factorisation of b. Then*

a) The power of p in ab is $d_a + d_b$.

b) The power of p in $\gcd(a, b)$ is $\min(d_a, d_b)$.

c) The power of p in $\operatorname{lcm}(a, b)$ is $\max(d_a, d_b)$.

Proof. We prove b). (The other parts can be an exercise.) Let d be the power of p in $\gcd(a, b)$. Then $p^d \mid \gcd(a, b)$. So $p^d \mid a$ and $p^d \mid b$. □

With this lemma, we are easily able to prove Theorem 4.23 which states that $\operatorname{lcm}(a, b) \cdot \gcd(a, b) = ab$.

Proof. For any prime p in the prime factorisation of ab the power of p in ab is $d_a + d_b = \min(d_a, d_b) + \max(d_a, d_b)$. On the other hand, the power of p in $\operatorname{lcm}(a, b) \cdot \gcd(a, b)$ is the power of p in $\operatorname{lcm}(a, b)$ plus the power in $\gcd(a, b)$, so is $\min(d_a, d_b) + \max(d_a, d_b)$.

This is true for any p dividing ab so the prime factorisations of ab and $\operatorname{lcm}(a, b) \cdot \gcd(a, b)$ are the same. We conclude that they are equal. □

Definition 4.28. Let $d(n)$ be the number of positive divisors of n.

Problem 4.8. Find $d(2^3 \cdot 3^5 \cdot 7^2)$.

Solution

Each positive divisor of $2^3 \cdot 3^5 \cdot 7^2$ is of the form $2^a \cdot 3^b \cdot 7^c$ where $a \in \{0, 1, 2, 3\}, b \in \{0, 1, \ldots, 5\}$, and $c \in \{0, 1, 2\}$. So there are $4 \times 6 \times 3$ different ones.

In a similar way one could show that where an integer n has prime decomposition

$$n = p_1^{d_1} \cdot p_2^{d^2} \cdot \ldots \cdot p_r^{d_r},$$

we have $d(n) = (d_1 + 1)(d_2 + 1) \ldots (d_r + 1)$.

4.6 Integers Modulo m (or clock numbers)

The following material is similar to that covered in Section 14.3 of the text, but fundamentally different. We are not defining a new number system here, simply a function of the integers.

For integers a and m, the division algorithm gives us unique integers q and r such that $a = qm + r$ with $0 \leq r \leq m - 1$. There is a lot that we can do only with the value r, throwing away the q. We denote this r by $a \mod m$, saying 'a is equal to r modulo m We write $a \equiv_m b$ to mean $(a \mod m) = (b \mod m)$. Equivalently,

Definition 4.29. For integers a, b and m, we write

$$a \equiv_m b$$

to mean that $m \mid a - b$. We say 'a is congruent to b modulo m'.

For example $7 \mod 4 = 3$, and $-9 \equiv_{10} 1 \equiv_{10} 11$.

The following alternate definition of $a \equiv_m b$ should be clear.

Fact 4.30. *We have $a \equiv_m b$ if and only if there is some integer k such that $a = b + mk$.*

Problem 4.9. Show that $4^n \equiv_9 3n + 1$ for all $n \geq 1$.

Solution

This is clear if $n = 1$, indeed $4^1 = 3(1) + 1$. Assume it has been proved for $n = k - 1$, that is , that $4^{k-1} \equiv_9 3(k-1) + 1 + 9c$ for some integer c. Then

$$\begin{aligned}
4^k &= 4(4^{k-1}) \\
&= 4(4^{k-1}) \\
&= 4\left((3(k-1)+1)+9c\right) \text{ by the induction hypothesis} \\
&= (3+1)(3(k-1)+1) + 4 \cdot 9c \\
&= 3(k-1) + 9(k-1) + 3 + 1 + 9(4c) \\
&= 3k + 1 + 9(k - 1 + 4c) \\
&\equiv_9 3k + 1
\end{aligned}$$

The following rules would simplify the above proof.

Theorem 4.31. *If $a \equiv_m b$ and $c \equiv_m d$, then*

i) $a + c \equiv_m b + d$

ii) $a - c \equiv_m b - d$

iii) $ac \equiv bd$

Proof. We do only ii). The other proofs are similar. As $a \equiv_m b$ and $c \equiv_m d$, we have by definition that $m \mid a - b$ and $m \mid c - d$. So $m \mid (a - b) - (c - d) = (a - c) - (b - d)$. By definition we get ii). □

Cancelation doesn't for work in general in the integers modulo m, because for example $2(1) \equiv_6 2(4)$ but $1 \not\equiv_6 4$. But it does work sometimes.

Theorem 4.32. *If $ac \equiv_m bc$ and $\gcd(c, m) = 1$, then $a \equiv_m b$.*

Proof. If $ac \equiv_m bc$ then $m \mid ac - bc = c(a-b)$, so $m \mid c$ or $m \mid (a-b)$. If we also have that $\gcd(c, m) = 1$, then $m \nmid c$, and so $m \mid a - b$. That is $a \equiv_m b$. $\square$

Problem 4.10. Show that 3 divide n if and only if 3 divides the sum of the digits of n.

Solution

Let $n = d_r(10^r) + d_{r-1}(10^{r-1} + \ldots d_0$. Since $10 \equiv_3 1$ we have

$$\begin{aligned} n &= d_r(10^r) + d_{r-1}(10^{r-1} + \ldots d_0 \\ &\equiv_3 d_r(1^r) + d_{r-1}(1^{r-1}) + \ldots d_0 \\ &= d_r + d_{r-1} + \ldots d_0 \end{aligned}$$

Problem 4.11. Find the last digit of 7^{100}.

Solution

$7^2 = 49 \equiv_{10} -1$, so $7^{100} \equiv_{10} -1^{50} = 1$.

Problem 4.12. Find $15(17) - 19 \mod 7$.

5 Computational Complexity

This chapter is Section 5.7 of the text

We look at how a function $f : \mathbb{Z} \to \mathbb{R}$ grows when $n \in \mathbb{Z}$ is large- what the graph of f looks like from far away.

Example 5.1. We would like so say approximately that

- $2n^2 - 3n + 7$ grows like n^2,
- $100\sqrt{n} + \ln n$ grows like $\sqrt{n}$, and
- $2^n - n!$ grows like $n!$.

This will be the statement that f is *of order* g. We start with a one sided version of this.

Definition 5.2. Let $f, g : \mathbb{Z} \to \mathbb{R}$ be functions. We write $f = O(g)$, and say that 'f is Big-oh of g' or 'f is dominated by g' or 'f is of order at most g' if there exists an integer k and a real number m such that for all $n \geq k$,

$$|f(n)| \leq m \cdot |g(n)|.$$

Problem 5.1. Show that $2n^2 - 3n + 7$ is $O(n^2)$.

Hint

See Example 5.65 of text.

Problem 5.2. Show that $2^n = n!$ is $O(n!)$.

These proofs should feel a lot like 'limits' from calculus. In fact, an alternate definition that f is $O(g)$ is that $|f(n)|/|g(n)| \leq m$ for large enough n. As such, a quick way to show that f is dominated by g is to show that

$$\lim_{n\to\infty} \frac{|f(n)|}{|g(n)|} \neq \infty.$$

Example 5.3. We show that $100\sqrt{n} + \ln n \in O(\sqrt{n})$. Indeed,

$$\begin{aligned}
\lim_{n\to\infty} \frac{|100\sqrt{n} + \ln n|}{|\sqrt{n}|} &= \lim_{n\to\infty} \frac{100\sqrt{n} + \ln n}{\sqrt{n}} \\
&= \lim_{n\to\infty} 100 + \frac{\ln n}{\sqrt{n}} \\
&\overset{l'hopital}{=} 100 + \lim_{n\to\infty} \frac{1/n}{n^{-1/2}} \\
&= 100 + \lim_{n\to\infty} \frac{\sqrt{n}}{n} = 100\sqrt{n}
\end{aligned}$$

So $100\sqrt{n} + \ln n < 101\sqrt{n}$ for large enough n.

This is no use, however in showing that f is not in $O(g)$:

$$f(n) = \begin{cases} 1 & \text{if } n \text{ is even} \\ 0 & \text{if } n \text{ is odd} \end{cases}$$

is $O(1)$ but $\lim_{n\to\infty} \frac{|f(n)|}{1}$ does not exist.

In fact, showing that f is not in $O(g)$ has a different flavour than showing it does. We have to negate the quantifications.

Problem 5.3. Show that $2^n - n!$ is not $O(2^n)$.

Hint

See Example 5.66 of text

Problem 5.4. Show that $2n^2 - 3n + 7 \notin O(n)$.

5.0.1 Lower Bounds

Definition 5.4. For functions $f, g : \mathbb{Z} \to \mathbb{R}$, f is $\Omega(g)$, read 'big omega', if there exist m, k such that

$$|f(n)| \geq m|g(n)| \forall n \geq k.$$

Alternately: f is $\Omega(g)$ if and only if g is $O(f)$. Indeed $|f| \geq m|g|$ if and only ig $|g| \leq 1/m|F|$.

Definition 5.5. If f is $O(g)$ and $\Omega(g)$ then it is $\Theta(g)$, read 'big theta of g'. Equivalently, f is $\Theta(g)$ if there are k, m_1, m_2 such that for all $n \geq m$

$$m_1|g(n)| \leq |f(n)| \leq m_2|g(n)|.$$

Equivalently, f is $\Theta(g)$ if and only if f is $O(g)$ and g is $O(f)$; we say f and g have the same order.

Problem 5.5. Show that $f(n) = 1^2 + 2^2 + 3^2 + \ldots n^2$ is $\Theta(n^3)$.

From the examples/problems of this section, we can infer a heirarchy of orders:

$$\Theta(1) < \Theta(\log n) < \Theta(n^{.02}) < \Theta(n) < \Theta(n^5) < \Theta(C^n) < \Theta(n!)$$

(where $C > 1$).

Problem 5.6. Find the order of

a) $3n + 7$

b) $3 + \sin(1/n)$

c) $3\sqrt{n}\log n + (\log n)^2$

d) $\log(\log n) + \log n$

10 Recurrence Relations

10.1 First Order Recurrence Relations

In this chapter we look at recursively defined sequences $(a_0, a_1, a_2, \dots,)$ of numbers. A *recurrence relation* is an equation, or rule, that defines the nth number a_n in terms of earlier numbers.

Example 10.1. The equation $a_n = 5a_{n-1}$ is a recurrence relation that is satsified by many sequences of numbers: $(0, 0, 0, \dots)$ is one. $(1, 5, 25, \dots,)$ and $(2, 10, 50, \dots,)$ are others.

Observe that upon choosing a first term a_0, (indeed any term in the example above,) the recurrence equation defined the whole sequence.

Example 10.2. Setting $a_0 = 3$ and $a_n = 5a_{n-1}$ for $n \geq 1$ defines the sequence $(a_0, a_1, a_2, \dots) = (3, 15, 75, \dots)$. One can show by induction that a_n has an explicit representation as $a_n = 3 \cdot 5^n$.

In general, the recurrence relation

$$a_0 = A, \qquad a_{n+1} = d \cdot a_n \text{ for all } n \geq 0$$

defines the sequence with explicit formula $a_n = Ad^n$ for all $n \leq 0$. Such an explicit formula is called the solution of a recurrence relation.

Problem 10.1. You invest \$1000 in the bank at .5% interest per month. How much is it worth after 2 years?

Solution

The value v_n in dollars after n months is determined by the initial value $v_0 = 1000$, and the recurrence relation $v_{n+1} = v_n \cdot 1.005$. So it is $v_{24} = 1000(1.005)^{24}$ dollars.

The above recurrence relations are quite simple. They are called *homogeneous first order linear recurrence relations with constanst coefficients*, and we have solved them all.

A recurrence relation that can be written in the form

$$c_0 a_n = c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k} + d$$

is called a kth *order linear* recurrence relation. If $d = 0$, it is called *homogeneous* and if the c_i are constant, it is said to have *constant coefficients.*

In later sections we solve all homogeneous second order linear recurrences with constant coefficients, and then some non-homogeneous recurrences. These take more theory. To finish off this section we look as some special cases that we can do without much more theory.

Problem 10.2. You invest \$100 per month in a bank at an interest rate of .5% per month. How much money do you have after one year?

Solution

This is the recurrence $a_n = 100 + (1.005)a_{n-1}$ with $a_0 = 100$ we solve this more generally in the next example.

Example 10.3. Solve the (non-homogeneous) recurrence relation defined by $a_0 = 0$ and $a_{n+1} = da_n + c$.

Solution

Writing out the first couple terms we have that $a_1 = c$, $a_2 = c(1 + d)$, $a_3 = c(1 + d + d^2)$. We can show by induction that this pattern continues and has the solution $a_n = c(1 + d + d^2 + \cdots + d^{n-1}) = c\frac{d^n-1}{d-1}$.

Example 10.4. Solve the (non-linear) recurrence relation defined by $a_0 = 5$, and $a_{n+1}^3 = 2a_n^3$.

Solution

It isn't linear in a_n, but making a change of variable $b_n = a_n^3$, the recurrence $b_0 = 5$ and $b_{n+1} = 2b_n$ is, and has solution $b_n = 5 \cdot 2^n$. So $a_n = (5 \cdot 2^n)^{1/3}$ is the solution to our original relation.

10.2 Second order linear homogeneous recurrence relations with constant coefficients

Second order linear homogeneous recurrence relations can be written as

$$c_0a_n + c_1a_{n-1} + c_2a_{n-2} = 0.$$

To solve this, we will guess a solution of the form $a_n = cr^n$ for non-zero c and r, and plug it in to the recurrence relation.

Now $c_0(cr^n) + c_1(cr^{n-1}) + c_2(cr^{n-2}) = 0$ if and only if $c_0r^2 + c_1r^1 + c_2 = 0$. This is only true if r is a root of the *characteristic equation* $c_0x^2 + c_1x + c_0 = 0$. Let's see how this plays out in an example.

Example 10.5. (Distinct Real Roots) Solve the recurrence relation $a_n = 5a_{n-1} - 6a_{n-2}$, $n \geq 2$, $a_0 = 1$, $a_1 = 5$.

Solution

1) Rewriting the recurrence relation as $a_n - 5a_{n-1} + 6 = 0$, the characteristic equation is $x^2 - 5x + 6 = 0$.

2) The characteristic equation has *characteristic roots* $r = 2, 3$.

3) Each of these is a solution to the recurrence relation. As it is linear, any linear combination $A(2^n) + B(3^n)$ is also a solution, so we have the *general solution*

$$a_n = A(2^n) + B(3^n)$$

(for constants A and B) to the recurrence $a_n = 5a_{n-1} - 6a_{n-2}$.

4) Using $a_0 = 1$ and $a_1 = 5$, we get a *specific solution*. From the first *initial condition* we get, $1 = a_0 = A(2^0) + B(3^0) = A + B$. From the second, we get $5 = a_1 = A(2) + B(3) = 2A + 3B$. Solving these, we get $B = 3$ and $A = -2$. So the specific solution to our recurrence relation is

$$a_n = (-2)2^n + (3)3^n = 3^{n+1} - 2^{n+1}.$$

Remark 10.6. It can be shown, using linear algebra, that the the general solution given in part 3) describes all possible solutions of the recurrence relation $a_n = 5a_{n-1} - 6a_{n-2}$. Indeed, the recurrence can be rewritten $a_n - 5a_{n-1} - 6a_{n-2} = 0$. As a system of equations in the variables a_n, a_{n-1}, and a_{n-2} the solution space has dimension 2. We won't prove this, we will just believe it.

This lays out a general process for solving linear homogeneous recurrence relations with constant coefficients.

i. Find the characteristic equation.

ii. Find the characteristic roots.

iii. Find the general solution.

iv. Use the initial conditions to find the specific solution.

Try this process to solve the following.

Problem 10.3. Solve the following (shifted fibonacci) recurrence relation: $f_n = f_{n-1} + f_{n-2}$, $f_0 = 0$, and $f_1 = 1$. (See Example 10.10 of the text book. You should get that $f_n = \frac{1}{\sqrt{5}}(\frac{1+\sqrt{5}}{2})^n - \frac{1}{\sqrt{5}}(\frac{1-\sqrt{5}}{2})^n$).

This process works for higher order relations than 2nd order, however we may have trouble finding the characteristic roots for higher order equations. We will mostly only consider 2nd order equations. The previous example was the easiest case. In general, our characteristic roots may not real. This is not a difficult complication. The more difficult complication is when they are real, but not distinct. (If they aren't real, the will be distinct.)

Example 10.7. (Complex Roots - Necessarily Distinct) Solve the recurrence relation $a_n = 2a_{n-1} - 2a_{n-2}$ with initial conditions $a_0 = a_1 = 1$.

Solution

There characteristic equation is $x^2-2x+2 = 0$, which has roots $x = 1\pm\sqrt{-1} = 1 \pm i$. So our general solution is

$$a_n = A(1+i)^n + B(1-i)^n.$$

We now find the specific solution. For one, we have $1 = a_0 = A + B$, so $B = 1 - A$. From the second initial condition, we have

$$1 = a_1 = A(1+i) + B(1-i) = A(1+i) + (1-A)(1-i) = 2Ai + 1 - i.$$

Solving gives $A = B = 1/2$, so we get the specific solution

$$a_n = \frac{1}{2}\left((1+i)^n + (1-i)^n\right).$$

Using our knowledge of complex numbers, we observe that $(1+i)^n$ and $(1-i)^n$ are conjugate. Thus their sum is twice their real part, which is $\sqrt{2}\cos(n\pi/4)$, and so we get that $a_n = \sqrt{2}\cos(n\pi/4)$. However, it isn't assumed that we know how to deal with complex numbers, so it's fine for you to leave the solution as $a_n = \frac{1}{2}\left((1+i)^n + (1-i)^n\right)$.

Example 10.8. (Repeated real roots) Solve $a_n = 6a_{n-1} - 9a_{n-2}$, where $a_0 = 3$ and $a_1 = 10$.

Solution

The characteristic equation $x^2 - 6x + 9 = (a-3)^2 = 0$ has double root 3. In this case, we have a solution $a_n = A3^n$, for all A, but this is not all of the solutions. Observe that $n3^n$ is also a solution:

$$n3^n = 6(n-1)3^{n-1} - 9(n-2)3^{n-2}.$$

Again linearity gives that

$$a_n = A3^n + Bn3^n$$

is a solution for all A and B. This is our general soluton.

Using the initial conditions, we can sovle to get the specific solution $a_n = 3^{n+1} + n3^{n-1}$.

Problem 10.4. Solve $a_n = -2a_{n-1} - a_{n-2}$ where $a_0 = 2$ and $a_1 = 5$.

Solution

You should find that the characteristic equation has a double root -1, and get a specific solution $(2-7n)(-1)^n$.

As long as we can find the characteristic roots, we should be able to solve higher order homogeneous linear equations with constant co-efficients.

Example 10.9. (Third Order) Solve $2a_{n+3} = a_{n+2} + 2a_{n+1} - a_n$, where $a_0 = 0, a_1 = 1$, and $a_2 = 2$.

Solution

The characteristic equation $2x^3 - x^2 - 2x + 1 = 0$ has roots $x = \pm 1, 1/2$. So the general solution is $a_n = A(1)^n + B(-1)^n + C(1/2)^n$. Now we find the particular solution. From the first initial condition, $0 = a_0 = A+B+C$, from the second $1 = a_0 = A-B+C/2$, from the third we have $2 = a_2 = A+B+C/4$. This is three equations in three unknowns, so we can solve it to get $A = 5/2$, $B = 1/6$, and $-8/3$. So $a_n = \frac{5}{2} + (\frac{-1}{6})^n - \frac{8}{3}(\frac{1}{2})^n$.

Now let's see a bit how we can use these recurrence relations.

Problem 10.5. How many binary strings (strings of '0's and '1's) of n bits have no two consecutive zeros?

Solution

Let a_n be the number of such strings. Clearly $a_0 = 0$, $a_1 = 2$, and $a_2 = 3$. An n bit string begins (1, and ends with any $n-1$ bit string, or begins with $(0, 1$, and ends with any $n-2$ bit string. So $a_n = a_{n-1} + a_{n-2}$. We can solve this.

This is almost the same problem.

Problem 10.6. I have buttons of four different colours. How many ways can we arrange n of them so that there are no two consecutive blue ones.

Solution

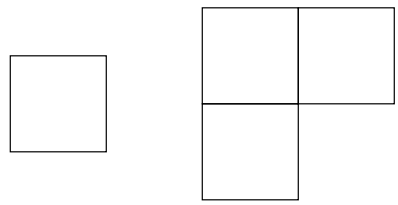
You should get a recurrence relation $a_n = 3a_{n-1} + 3a_{n-2}$.

Problem 10.7. How many compositions of the integer n use only the digits 1 and 2?

Solution

We add a 1 to the end of any composition of $n-1$ and a 2 to the end of any composition of $n-2$. So this is described by the recurrence relation $a_n = a_{n-1} + a_{n-2}$. The initial conditions are $a_1 = 1$ and $a_2 = 1$. Fibonacci fabulous!.

Problem 10.8. How many ways can you tile a $2 \times n$ checkerboard with the



tiles of the type seen here.

Solution

This is Example 10.19 from the text.

10.3 The Nonhomogeneous Recurrence Relation

A linear recurrence relation, written as

$$a_n + c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_d a_{n-d} = f(n)$$

is called *nonhomogeneous* if $f(n)$ is not identically 0.

We will look at the first and second order cases. Again, one can show by linear algebra that to find the general solution to such a relation, one only needs to find the general solution $a_n^{(h)}$ to a *corresponding homogeneous recurrence relation*, and then shift it by one *particular solution* $a_n^{(p)}$.

We start with an example.

Example 10.10. Solve $a_n = 2a_{n-1} + 7 \cdot 5^n$ for $n \geq 1$ and $a_0 = 4$.

Solution

First we find the general solution to the corresponding homogeneous recurrence $a_n - 2a_{n-1} = 0$. This has characteristic equation $x - 2 = 0$ which has root $x = 2$ so we have the general soluton $a_n^{(h)} = A \cdot 2^n$.

Now we want a particular solution to $a_n - 2a_{n-1} = 7 \cdot 5^n$. Lets guess that there is one of the form $c \cdot 5^n$. (We talk about why after this example.) Plugging into the recurrence relation, we get

$$c \cdot 5^n - 2c \cdot 5^{n-1} = 7 \cdot 5^n,$$

which is true if and only if $c \cdot 5 - 2c = 35$, or $c = 35/3$. So $a_n^{(p)} = (35/3) \cdot 5^n$ is a solution to the recurrence.

The general solution to the recurrence is thus

$$a_n = a_n^{(h)} + a_n^{(p)} = A \cdot 2^n + (35/3) \cdot 5^n.$$

Using the initial conditions we now find the specific solution. We have $4 = a_0 = A(2^0) + (35/3)5^0 = A + 35/3$, which gives $A = -23/3$. The specific solution is thus

$$a_n = (35/3)5^n - (23/3)2^n = (1/3)(35 \cdot 5^n - 23 \cdot 2^n).$$

For $f(n) = 7 \cdot 5^n$ we guessed a solution of the form $c \cdot 5^n$. The solution that we guess is suggested by the form of $f(n)$:

When $f(n)$ is	$a_n^{(p)}$ is of the form
c	A
cn	$An + B$
cn^2	$An^2 + Bn + C$
cr^n	Ar^n
cn^2r^n	$r^n(An^2 + Bn + C)$

From the above, you can interpolate appropriate guesses for various other special cases of $f(n)$. You will be responsible only for solving nonhomogeneous equations with $f(n)$ from the above table.

Problem 10.9. Solve $a_n = -6a_{n-1} - 9a_{n-2} + (-3)^n$ for $n \geq 2$, where $a_0 = 2$ and $a_1 = 3$.

Example 10.11. (The Towers of Hanoi) In the game of the Towers of Hanoi one has three pegs. A stack of disks of different radii are stacked on one peg with no disk on top of a larger disk. The game is to move the stack of disks to another peg. In a move, one can move one disk, placing it on an empty peg, or on a peg of smaller radius. How many moves can this be done in when there are n disks.

Solution

Let a_n be the minimum number of moves one can complete the game in when there are n disks. Clearly $a_1 = 1$. To move n disks one must move the largest disk. When this is done, it must be moved from peg 1 to peg 3, so at this time all other disks must be stacked on peg 2. Up until this move, there must be at least a_{n-1} moves, and after the biggest disk is moved, we must move these other disks on top of it. This takes a_{n-1} moves. So $a_n = 2a_{n-1}+1$. We have a recurrence relation which we can solve.

The corresponding homogeneous equation has characteristic polynomial $x - 2 = 0$ so general solution $a_n^{(h)} = A \cdot 2^n$. The nonhomogeneous portion of the recurrence relation is 1 which suggests a particular solution of the form B. Plugging in, we get $B = 2B + 1$ which we solve for the particular solution $a_n^{(p)} = -1$. So the nonhomogeneous recurrence has general solution $a_n^{(h)} = A \cdot 2^n - 1$.

Using the initial condition that $1 = a_1 = A \cdot 2 - 1$, we get $A = 1$, so the specific solution is thus $a_n = 2^n - 1$.

The best one can do for the n disk game is $2^n - 1$ moves.

11 Graph Theory

This is from Chapter 11 of the text.

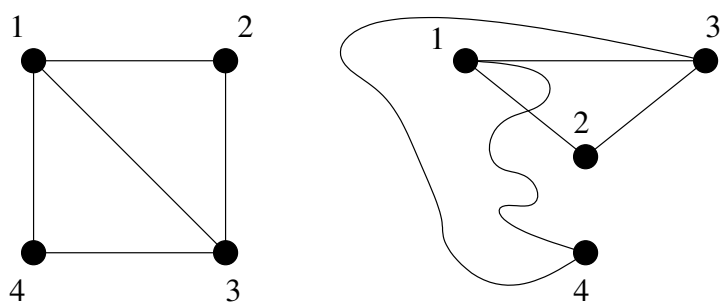
11.1 Introduction to Graph Theory

Definition 11.1. A graph $G = (V, E)$ consists of a non-empty set V of *vertices* and a set E of 2-element subsets of V, called *edges*.

For example

$$G = (\{1, 2, 3, 4\}, \{\{1, 2\}, \{2, 3\}, \{3, 4\}, \{4, 1\}, \{1, 3\}\})$$

is a graph. It has four vertices and five edges. As this mess of brackets gets troublesome to write, we drop them when it doesn't confuse things an write an edge $\{x, y\}$ as xy. The graph G can be represented pictorially as



We often write $V(G)$ for V and $E(G)$ for E if we don't define the these sets for a graph G explicitly.

There are a lot of definitions related to the fact that $e = uv$ is an edge of G. We say

- the vertices u and v are *adjacent*, written $u \sim v$,
- u and v are *neighbours*,
- u and v are the *endpoints* of e,
- u (and v) is incident with e,

Note

One can generalise the idea of a graph to

- *digraphs* by making edges ordered sets: uv and vu are different,
- *multigraphs* by making the edgeset E a multiset,
- *hypergraphs* by allowing edges to contain more than two vertices,
- *non-simple graphs* by allowing edges of the form uu.

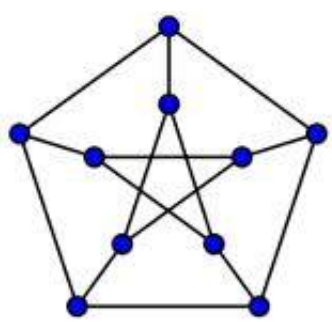
We mostly don't consider these generalisations.

We used graphs when we considered the problem of Ramsey:

> How many people must there be at a party so that we can be sure that there are three mutual acquaintences, or three mutual strangers?

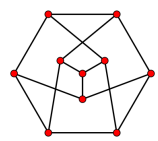
We represented the people at the party as vertices of a graph, and the relationship of 'begin acquainted' by edges. This is just one example of how graphs can be used to model problems. Graph theory studies the properties that arise in structure due to the simple idea of a relationship between elements of a set. Many such properties are purely combinatorial, others have more structural ramifications, which arise surpisingly in most any field of mathematics.

Here is a graph, called the *Petersen Graph*, and denoted P, that some people like to put on t-shirts, and refer to fondly as the *universal counter-example*:



Here are some typical questions that we might ask of this, or any graph. The definitions you need to understand these questions come later, but they are probably what you think they are.

- How many colours do we need, when colouring the vertices of P to be sure that adjacent vertices get different colours.

- Can you draw P without intersecting edges?
- How many edges (vertices) must you remove to *disconnect* P?
- For any two pairs of vertices u, u' and v, v' can you find a *path* between u and u', and another between v and v', such that the two are disjoint?
- What is the smallest *cycle* in P?
- How many 5-cycles are there in P?
- Can we partition the edges of P into disjoint 5-cycles?
- Can we find a *walk* in P that visits every vertex (edge) exactly once?
- Is P the same graph as this: 

11.1.1 Isomorphisms

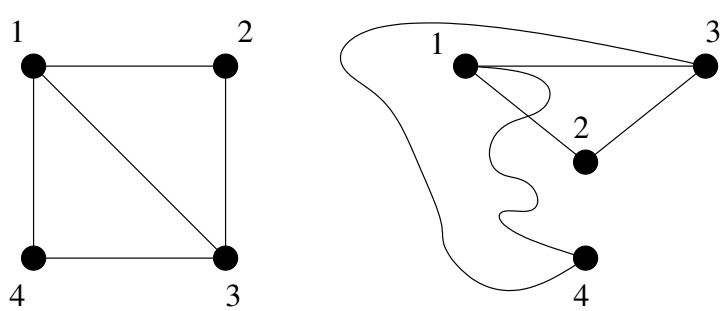
Often we don't care about the names of the vertices, and want to consider graphs to be the same if they look the same when ignoring these vertex names. For this we make the following defintion.

Definition 11.2. An *isomorphism* is a bijective map $f : V(G) \to V(H)$ such that

$$uv \in E(G) \iff f(u)f(v) \in E(H).$$

If there is an isomorphism f from G to H, we say G and H are *isomorphic*, and write $G \cong H$.

Example 11.3. The following graphs are all isomorphic.



It is a hard problem to decide if two graphs are isomorphic. We talk about this a little more later.

Theorem 11.4. *Isomorphism is an equivalence relation. That is:*

i. $G \equiv G$ *(symmetry)*

ii. $G \equiv H \Rightarrow H \equiv G$ *(reflexivity)*

iii. $G \equiv H$ *and* $H \equiv J \Rightarrow G \equiv J$ *(transitivity)*

Problem 11.1. Prove this!

Problem 11.2. How many graphs are there on 4 vertices? How many non-isomorphic (unlabelled) graphs?

As we usually consider graphs to be the same if they are isomorphic, we would like to decide if two graphs are isomorphic. Deciding if two graphs are isomorphic is usually pretty hard (this statement can be made rigourous). But if we are lucky, there are easy ways to prove that they are not.

For example, if two graphs have a different number of vertices, or a different number of edges, they are not isomorphic. Such properties, which are 'preserved by isomorphism', are called graph properties. Graph Theory is the study of these properties.

11.1.2 Some Common Graphs

We generally now consider graphs to be the same if they are isomorphic, and take blatent liberties with the names of their vertices. As such, we define some common graphs, but will apply these names to any graph that is isomorphic.

- The complete graph on n vertices, K_n, has $V(K_n) = \{v_1, v_2, \dots v_n\}$ and $E(K_n) = \{v_i v_j \mid 1 \leq i < j \leq n\}$.
- The n-cycle, C_n, has $V(C_n) = \{v_1, v_2, \dots v_n\}$ and $E(C_n) = \{v_i v_{i+1} \mid i = 1, \dots, n-1\} \cup \{v_n v_1\}$.
- The n-path, P_n, has $V(P_n) = \{v_0, v_1, \dots v_n\}$ and $E(P_n) = \{v_i v_{i+1} \mid i = 0, \dots, n-1\}$.
- The complete bipartite graph, $K_{m,n}$, has $V(K_{m,n}) = \{a_1, \dots a_m\} \cup \{b_1, \dots, b_n\}$ and $E(K_{m,n}) = \{a_i b_j \mid i \in [m], j \in [n]\}$.
- The n-cube, Q_n, $V(Q_n)$ is the set of binary strings of length n, vertices u and v are adjacent if they differ in exactly one coordinate.

11.1.3 Homomorphisms

Definition 11.5. A *homomorphism* $f : G \to H$ of a graph G to another graph H, is a vertex map $f : V(G) \to V(H)$ such that

$$uv \in E(G) \Rightarrow f(u)f(v) \in E(H).$$

If there is a homomorphism $f : G \to H$, we write simply $G \to H$.

Any isomorphism is a homomorphism, and as usual, a homomorphism $f : G \to H$ is an isomorphism if and only if there is an inverse homomorphism $f^{-1} : H \to G$ such that $f \circ f^{-1}$ and $f^{-1} \circ f$ are the identity maps. But there is another property you may be used to for other finite structures, that doesn't hold for us: **a bijective homomorphism is not necessarily an isomorphism.** Indeed, there are many bijections that are not isomorphisms.

Example 11.6. For any graph G on n vertices there is a bijective homomorphism $f : G \to K_n$.

A homomorphism $f : G \to H$ is a map on the vertices, but it induces a map $f_E : E(G) \to E(H)$ on the edges:

$$f_E(uv) = f(u)f(v).$$

The homomorphism f is called *edge-injective* if f_E is injective, and *edge-surjective* if f_E is surjective.

Problem 11.3. Show that if f is injective, then it is edge-injective, but that the converse is not true.

Problem 11.4. Show that f is an isomorphism if and only if it is bijective and edge-surjective.

Problem 11.5. Show that $C_{2n+1} \to C_{2m+1}$ if and only if $m \leq n$. Show that $C_n \to K_2$ if and only if n is even. Show that $G \to K_2 \Rightarrow C_{2n+1} \not\to G$ for every n.

11.2 Subgraphs, Complements, and Degree Sequences

Given a graph G, another graph G' is a *subgraph* of G, written $G' \leq G$ if $V(G') \subset V(G)$ and $E(G') \subset E(G)$.

A subgraph G' of G is *spanning* if $V(G') = V(G)$, or *induced* if $E(G') = \{uv \in E(G) | u, v \in V(G')\}$. A subgraph G' of G is a *proper* subgraph of G if $E(G')$ is a proper subset of $E(G)$.

Problem 11.6. How many non-labelled subgraphs are there of K_4? How many are proper? ...induced? ...spanning? ...spanning, induced and proper?

Definition 11.7. The *complement* $\overline{G}$ of a graph G is the graph defined by $V(\overline{G}) = V(G)$ and $E(\overline{G}) = \overline{E(G)}$, (where the universe used for complementation is the set $\{uv \mid u \neq v \in V(G)\}$ of all possible edges on $V(G)$).

Problem 11.7. Show that a subgraph G' of G is induced if and only if $\overline{G'}$ is a subgraph of $\overline{G}$.

Solution

Let $G' \subset G$ be induced. Clearly $V(\overline{G'}) \subset V(\overline{G})$, so we have to show that $E(\overline{G'}) \subset E(\overline{G})$. Let $e \in E(\overline{G'})$ then $e \notin E(G')$ so $e \notin E(G)$. So $e \in E(\overline{G})$.
On the other hand, let $\overline{G'} \leq \overline{G}$. We want to show that G' is induced. For $u, v \in V(G')$, with $uv \in E(G)$ we want to show that $uv \in E(G')$. Well $uv \in E(G)$ implies that $uv \notin E(\overline{G})$ so not in $E(\overline{G'})$, so because $u, v \in V(G')$, $uv \in E(G)$.

Problem 11.8. Show that a bijective homomorphism $f : G \to H$ is an isomorphism if and only if it is also a homomorphism of $\overline{G} \to \overline{H}$.

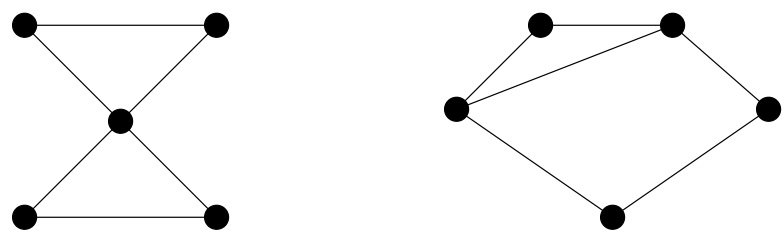
Problem 11.9. A graph G is self-complementary if $G \equiv \overline{G}$. How many edges are there in a self-complementary graph on n vertices?

11.2.1 Degrees and Degree Sequences

Definition 11.8. For a vertex v in a graph G, the *degree* of v, written $\deg(v)$, is the number of edges in G incident to v. The *degree sequence* of a graph G is the multiset of the degrees of the vertices of G. The minimum degree, $\delta(G)$, and maximum degree, $\Delta(G)$, of a graph G are the minimum and maximum degrees respectively in its degree sequence.

Problem 11.10. Show that every graph contains a path of length at least $\delta(G)$.

Clearly, degree sequence, minimum degree, and maximum degrees are graph properties. So the following two graphs, which have the same number of vertices, and edges are not isomorphic as they have different degree sequences: $(4, 2, 2, 2, 2)$ and $(3, 3, 2, 2, 2)$ respectively, (and in fact different maximum degrees.)



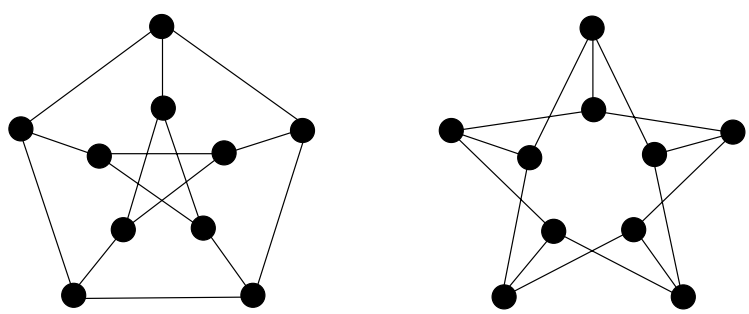
Here is a simple observation about degree sequence:

Theorem 11.9. *If $(d_1, d_2, \ldots, d_n)$ is the degree sequence of a graph, then the degree sum: $\sum_{i=1}^{n} d_i$ is equal to $2|E(G)|$.*

Proof. Each edge of the graph contributes 2 to the degree sum, one for each endpoint. □

Corollary 11.10. *In any graph G, there is an even number of vertices of odd degree.*

Problem 11.11. What are the degree sequences of the following graphs? Are the following two graphs, with the same degree sequences, isomorphic?



Problem 11.12. What is the degree sequence of the cube Q_n?

Definition 11.11. A graph G is called *k-regular* if every vertex has degree k.

Problem 11.13. How many edges does a k-regular graph on n vertices have?

A sequence $(d_1, d_2, \ldots, d_n)$ non-increasing integers is *graphic* if it is the degree sequence of a graph. We saw above that one necessary condition for a sequence

to be graphic is that it sums to an even number. But this is not sufficient: $(4, 2, 0)$ is not graphic.

Problem 11.14. Show that the following condition is also necessary for a sequence $(d_1, d_2, \ldots, d_n)$ to be graphic. For all $1 \leq k \leq n$,

$$\sum_{i=1}^{k} d_i \leq k(k-1) + \sum_{i=k+1}^{n} \min(k, d_i).$$

In fact, this condition is also sufficient. But this is fair bit harder to show:

Theorem 11.12. *(Erdős-Gallai '60) A sequence $(d_1, d_2, \ldots, d_n)$ is graphic if and only if $\sum_{i=1}^{k} d_i$ is even and for all $1 \leq k \leq n$,*

$$\sum_{i=1}^{n} d_i \leq k(k-1) + \sum_{i=k+1}^{n} \min(k, d_i).$$

The Erdős-Gallai condition is what we use to decide if a sequence is graphic, but there is also a recursive method to decide it. It depends on the following idea:

Lemma 11.13. *(Havel-Hakimi) If a sequence $D = (d_1, d_2, \ldots, d_n)$ is graphic, then there is a graph with degree seqence D such that the maximum degree vertex is adjacent to the d_1 vertices of next highest degree.*

Thus to decide if a sequence D, such as $(3, 3, 2, 2, 2)$, is graphic, the *Havel-Hakimi algorithm* is as follows. While D is not all zeros,

- remove d_1, and
- remove 1 from the next d_1 degrees $d_2, \ldots, d_{d_1-1}$,

if possible.

The algorithm clearly terminates. If it terminates with an empty sequence, (or a sequence of zeros), then D is graphic. Otherwise, it is not graphic.

Starting with $(3, 3, 2, 2, 2)$, we get $(\cancel{3}, 3-1, 2-1, 2-1, 2) = (2, 2, 1, 1)$, and then $(1, 0, 1) = (1, 1)$, and then $(0, 0)$. So $(3, 3, 2, 2, 2)$ is graphic.

After one definition, let's prove the Lemma.

Definition 11.14. The *neighbourhood* $N(v)$ of a vertex v in a graph G is the set

$$\{u \in V(G) \mid uv \in E(G)\}$$

of all neighbours of v.

Proof. Let $D = (d_1, d_2, \ldots, d_n)$ be a graphic sequence. Given a realisation G of D, let v be a vertex of degree $\Delta = d_1$, and let $T = (v_2, \ldots, v_{\Delta-1})$ be a set of Δ vertices where v_i has degree d_i for all i. If $N(v) = T$ then we are done, so towards contradiction, assume that the realisation G of D that maximises $\Delta' = |N(v) \cap T|$ has $\Delta' < \Delta$, and assume that G is such a realisation.

We find a new realisation G' with $|N(v) \cap T| > \Delta'$, so the lemma follows by contradiction.

As $N(v) \neq T$, there is $t \in T$ with $v \not\sim t$ and $s \notin T$ with $v \sim s$. Furthermore, since $t \in T$ and $s \notin T \cup \{v\}$, $\deg(t) \geq \deg(s)$. So there is some vertex w with $w \sim t$ and $w \not\sim s$.

Removing the edges vs and tw from G and adding vt and sw we get a graph G' with the same degrees on every vertex, (so another realisation of D), but with $|N_{G'}(v) \cap T| = |(N_G(v) \cap T) \cup \{t\}| = \Delta' + 1$, as needed for the contradiction. $\square$

Problem 11.15. Decide, using Havel-Hakimi, whether or not the sequence $(5, 5, 5, 3, 2, 2, 1, 1)$ is graphic.

11.3 Walks, trails, paths, circuits and cycles

Definition 11.15. Let x and y be vertices in a graph G. An xy-walk is a finite alternating sequence of vertices and edges:

$$x = x_0, e_1, x_1, e_2, e_2, \ldots, x_{n-1}, e_n, x_n = y$$

where $e_i = x_{i-1}x_i$ for $i = 1, \ldots n$.

Usually we just write the vertices

$$x_0 x_1 \ldots x_n,$$

or, if x and y are known, the edges

$$e_1 e_2 \ldots e_n.$$

A walk in a graph G can be viewed as a homomorphism of a path P_n to G. The *length* of the walk is n, the number of edges in it. If a walk is edge-injective, it is called a *trail*; if it is injective, it is called a *path*.

An xy-walk is *closed*, or a *circuit* if $x = y$. This can be viewed as a homorphism of a cycle C_n to G. If it is injective, it is called a *cycle*. The *girth* of the cycle is n.

Theorem 11.16. *Any xy-trail in a graph has an xy-path as a subgraph.*

Intuitively, we just start with an xy-trail an 'chop' out the repeated bits until it is a path. A lot of times it is very hard to write the intuitive proof rigorously. Try it before you proceed. A nice way to prove this theorem is the following proof by contradiction, using a minimum counterexample.

Proof. By assumption, there exists an xy trail in G. Let $x = x_0, x_1, \dots x_d = y$ be the shortest xy trail.

Towards contradiction, assume that the trail isn't a path. Then there exist some $0 \leq i < j \leq d$ such that $x_i = x_j$. But then

$$x_0, x_1, \dots, x_{i-1}, x_i, x_{j+1}, \dots, x_d$$

is an xy trail that is shorter than the original one. This is a contradiction, so this path is a trail. □

Problem 11.16. Prove the Theorem 11.16 using induction.

Problem 11.17. Prove the following. Any odd circuit in a graph has an cycle as a subgraph.

Definition 11.17. The *girth* of a graph is the minimum girth of all cycles in the graph.

Problem 11.18. What is the girth of the cube Q_n? ..of the Peteresen graph?

Problem 11.19. Show that for any vertex u and any other vertex v of the Petersen graph P, there is an *automorphism* f (that is, an isomorphism $f : P \to P$) such that $f(u) = v$. Graphs with this property are called *vertex transitive.*

11.3.1 Distance

Definition 11.18. The *distance*, $\mathrm{dist}(u, v)$ between two vertices u and v in a graph G is the length of the shortest uv-walk in G. The *diameter* $\mathrm{diam}(G)$ of G is the maximum $\mathrm{dist}(u, v)$ over all pairs u and v of vertices in G. The *radius* $\mathrm{rad}(G)$ of G is

$$\min_{u \in V(G)} \max_{v \in V(G)} \mathrm{dist}(u, v).$$

A vertex u that minimises $\max_{v \in V(G)} \mathrm{dist}(u, v)$ is called central.

Problem 11.20. What is the diameter of K_n, C_n, Q_n?

Problem 11.21. Show that for a graph G containing a cycle, the girth of G is at most $2\mathrm{diam}(G) + 1$.

Problem 11.22. How many vertices can a graph of radius k and maximum degree Δ have?

11.3.2 Connected Graphs

Definition 11.19. A graph G is called *connected* if there is an xy-path for every two vertices $x, y \in V(G)$.

Any *maximal* connected subgraph of G, that is, a connected subgraph that is not a proper subgraph of any other subgraph of G, is called a component of G.

Note

The word 'maximal' is different from 'maximum' and is not really used outside of mathematics. The word maximal is defined with respect to a *partial ordering*. (See Section 7.3 of the text for more about partial orders.) An element in a partial ordered set is *maximal* if it is greater than all elements with which it is comparible.
For graphs there are several different partial orders that we might talk about consider.

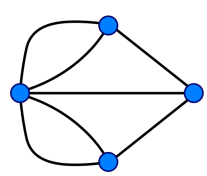
Generally, when we say 'maximal' we are talking about the partial order 'subset inclusion' a graph is maximal in a family of graphs if it is not a subgraph of any graph in the family but itself.

Example 11.20. Let G be the graph containing a K_4 on the vertices $\{1, 2, 3, 4\}$ and a P_2 on the vertices $\{a, b, c\}$. G has two components, K_4 and P_2.

11.3.3 Eulerian Graphs

It is often told that the first problem of graph theory was the problem of the Bridges of Königsberg. In the fictional city of Königsberg (Problem: prove or disprove my unpopular claim that Königsberg is fictional.) there were seven bridges. The dandies of the city, on slow Sunday, would walk about the town, playing a riddlish game. Starting whereever they pleased, they were to walk around, crossing every bridge, without crossing any bridge more then once. This game went on for centuries, and several liars claimed to have done it. Fermat once wrote in the margin of a book that it was simple enough. Until eventually, Euler proved it was impossible, and stabbed Fermat in a dual. (Prove that any of this is true.)

The city of Königsburg consisted of four dots, connected by seven bridges. It looked a lot like this:



How Euler proved it is lost in the depths of historical fiction. But our story inspires the following definitions.

Definition 11.21. An edge-bijective trail in G is called an *euler trail*; an edge-bijective circuit is called an *euler circuit*. A graph is called *eulerian* if it contains an euler circuit.

It is clear that a graph that is disconnected, with edges in each component, cannot be eulerian.

The following is what Euler claims to have proved.

Theorem 11.22. *A connected graph is eulerian if and only if all of its vertices have even degree.*

Proof. That an eulerian graph has all vertices of even degree is clear. Let $C = v_1 v_2 \dots v_N$ be an eulerian circuit. For any vertex v, count the number of times d that v occurs in $v_1 v_2 \dots v_{N-1}$. For each occurence $v = v_i$ is in two edges in C: $v_{i-1}v$ and vv_{i+1}. So v is in $2d$ edges in C. But C is edge-bijectve, so v is in $2d$ edges in G.

Now, in the other direction, assume that G is a connected graph in which every vertex has even degree. We must show that there is an euler circuit in G. There is an edge-bijective trail, indeed any length 1 trail is edge injective, so let $T = v_1 v_2 \dots v_m$ be a longest edge-injective trail in G. First observe that T must be closed. Indeed, if it weren't, then the degree of v_m in G is odd by our proof above, so there is another edge in G incident to $T(v_m)$. Adding this to T, we would get a longer edge-injective trail. So T is an edge-injective circuit.

If T is not edge-bijective, then $G' = G \setminus T$ is non-empty. As G is connected, some edge $e = u_1 u_2 \in G'$ has at least one endpoint $u_1 = v_i$ in $V(T)$. Letting $T' = u_1 u_2 \dots u_{m'}$ be a maximal trail in G' that begins with $u_1 u_2$ we again get that T' must be closed; that is $u_{m'} = u_1 = v_i$. But then

$$v_1 v_2 \dots, v_i u_1 u_2 \dots u_{m'} v_{i+1} \dots v_m$$

is an injective trail in G that is longer than T. This is a contradiction, so T must have been edge-bijective. Thus G is eulerian. □

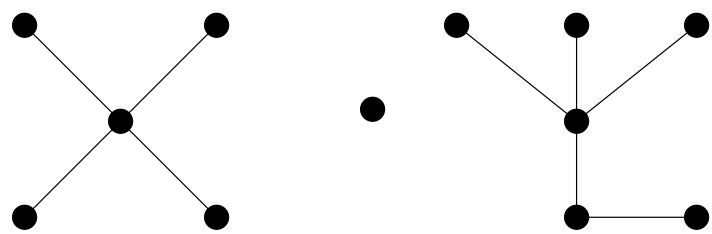
Problem 11.23. Give necessary and sufficient conditions for a graph to have an euler trail. Does Königsberg satisfy these conditions?

11.4 Trees

This is from Section 12.1 of the text. I like it better here.

Definition 11.23. A *forest* is a graph with no cycles as subgraphs. A *tree* is a connected forest.

The following graph is a forest, each of its three components is a tree.



Definition 11.24. A *spanning tree* of a graph G is a spanning subgraph which is a tree.

Theorem 11.25. *A graph G is connected if and only if it has a spanning tree.*

Proof. Let G be a graph with a spanning tree T. Let x and y be any two vertices in G. As T is spanning, they are also in T. As T is connected, there is an xy-path $P = e_1e_2 \dots e_d$ in T. But $E(T) \subset E(G)$ so P is in G as well, that is, it is an xy-path in G. This ways for any $x, y \in V(G)$, so G is connected.

On the other hand assume that G is connected. Let H be a minimal connected spanning subgraph of G, that is, a connected spanning subgraph, which itself has no connected spanning proper subgraph. We show by contradiction that H is a tree. Assume that it isn't. Then it contains a cycle $v_1v_2 \dots v_d$. Let $H' = H \setminus v_1v_2$. If H' is disconnected, then there exist vertices x and y of H such that there are no xy-paths in H'. So all xy-paths in H use v_1v_2, that is, they are of the form $x = x_0x_1 \dots x_iv_1v_2x_{i+1} \dots x_m = y$. But then $x_0x_1 \dots x_iv_1v_nv_{n-1} \dots v_2x_{i+1} \dots x_m$ is an xy-walk in H', contradicting the fact that there is no xy-path in H'. Thus H' is a smaller connected spanning subgraph, contradicting the choice of H. So by contradiction H is a tree. □

Definition 11.26. A *leaf* or *pendant edge* of a tree is a vertex of degree one.

Lemma 11.27. *Any tree has a leaf.*

Proof. Let G be a graph with no leaves. We show that G has a cycle, so is not a tree. This gives the contrapositive of the Theorem. Let $P = v_1v_2 \dots v_n$ be a longest path in G. As v_1 has some neighbour other than v_2 and there is no longer path than P, v_1 has a neighbour v_d on P, so $v_1v_2, \dots v_dv_1$ is a cycle in G. □

Theorem 11.28. *For a tree T, $|V(T)| = |E(T)| + 1$.*

Proof. The proof is by induction on $|V(T)|$. Since the only tree on one vertex is K_1 and the theorem is true for K_1 we assume that $n \geq 2$, and that the theorem is true for graphs on less than n vertices.

As T is a tree, it has a leaf v. Let $T' = T \setminus v$, the graph we get from T by removing v.

Claim 11.29. *T' is a tree*

Proof of claim. To see that it is connected, observe that the only paths in T that contain v, contain v as an endpoint. So for any vertices $x, y \in V(T) \setminus \{v\}$, the xy-path connecting them in T is also in T'. As $T' \leq T$ it has no cycles. So T' is a tree. $\diamond$

Now $|V(T)| - 1 = |V(T')|$, which by induction is $|E(T')| + 1 = |E(T)|$. So $|V(T)| = |E(T)| + 1$ as needed. $\square$

Definition 11.30. A *cut edge* of a graph G is any edge whose removal increases the number of components of G.

Theorem 11.31. *Let $G = (V, E)$ be a graph. Then the following are equivalent.*

i. G is a tree.

ii. G is connected, but every edge is a cut edge.

iii. G has no cycles, and $|V| = |E| + 1$.

iv. G is connected, and $|V| = |E| + 1$.

v. Any two vertices of G are connected by a unique path.

Proof. From the previous theorem we have that (i) implies (iii) and (iv). We show that (iii) $\Rightarrow$ (i) and (iv) $\Rightarrow$ (ii) $\Rightarrow$ (i). This shows the equivalence of the first four statements. As an exercise, show that statement (v) is also equivalent.

(iii) $\Rightarrow$ (i): Assume that G has no cycles, so each component of G is a tree. For each component C we thus have that $|V(C)| = |E(C)| + 1$. So $|V(G)| = |V(E)| + c$ where c is the number of components. By $|V| = |E| + 1$, there is only one component, so it is a tree.

(iv) $\Rightarrow$ (ii): Assume that G is connected an has $|V| = |E| + 1$. If there is some edge e that is not a cut-edge, then $G' = G \setminus \{e\}$, the graph we get by removing e, is connected, and so contains a spanning tree T. Thus $|V| = |V(G')| = |V(T)| = |E(T)| + 1 \leq |E(G')| + 1 = |E| + 2$, but this is a contradiction.

(ii) $\Rightarrow$ (i): Let G be connected and every edge of G be a cut-edge. As G is connected it has a spanning tree. Since every edge is a cut edge, no proper subgraph has a spanning tree, so G itself is a spanning tree.

□

11.5 Bipartite and Planar Graphs

11.5.1 Bipartite graphs

Definition 11.32. A graph G is *bipartite* if and only if there is a partition $A \cup B = V(G)$ of its vertices (so $A \cap B = \emptyset$), such that all edges are between A and B, (ie, that A and B both induce empty sets.

Problem 11.24. Show that if G is bipartite if and only if $G \to K_2$. Show that $C_{2n+1} \to G$ implies that G is not bipartite.

We can say more

Proposition 11.33. *For a graph G, the following are equivalent*

i. *G is bipartite.*

ii. *$G \to K_2$.*

iii. *G contains no odd cycles.*

iv. *$C_{2n+1} \not\to G$ for all integers $n \geq 1$*

Proof. We've proved most of this over the course of these notes. With impliction iii) $\Rightarrow$ ii), the proof can be pieced together from the exercises. So we prove this implication.

Assume that G has no odd cycles. We define a homomorphism $f : G \to K_2$ as follows. Let v_0 be some vertex in G. For every vertex v in G let

$$f(v) = \begin{cases} 0 \text{ if } \mathrm{dist}(v, v_0) \text{ is even} \\ 1 \text{ if } \mathrm{dist}(v, v_0) \text{ is odd} \end{cases}$$

We claim that this is a homomorphism. If it is not, there are some two vertices adjacent u and v such that $f(u) = f(v)$. That is, $\mathrm{dist}(u, v_0)$ and $\mathrm{dist}(v, v_0)$ have the same parity. But then there is path P_1 from v_0 to v and P_2 from u to v_0 of the same parity, so the path $P_1 + vu + P_2$ is an odd path from v_0 to v_0. That is, it is an odd circuit in G. But we showed this implies there is an odd cycle in G, which contradicts our assumption. □

Problem 11.25. Let G be a graph with n vertices and $m > (n/2)^2$ edges. Show that G cannot be bipartite.

11.5.2 Planar Graphs

Can we draw the petersen graph in the plane so that no edges cross?

A graph G is *planar* if it can be drawn in the plane with no edges crossing. A drawing of the graph in the plane with no edges crossing is called an *(planar) embedding* of G.

Problem 11.26. Show that K_4 is planar but that K_5 is not.

We can show that K_5 is not planar by ad-hoc arguments, but such arguments difficult for larger graphs. That's okay though. That K_5 is not planar tells us that a lot of other graphs are not planar.

Problem 11.27. Show that no graph containing K_5 as a subgraph, can be planar. Show that the petersen graph is not planar.

Definition 11.34. A *subdivision* of a graph G is a graph we get from G by dividing edges. That is, G' is a subdivision of G if we can construct it by repeated application of the following alteration, called an *edge subdivision*.

- remove an edge $e = ab$,
- add a new vertex v_e,
- add the edges av_e and bv_e.

It should be pretty clear that because K_5 is not planar, no subdivision of K_5 is planar. So no graph containing a subdivision of K_5 as a subgraph is planar. Can you think of any other graphs that are not planar? Please don't look ahead. It will ruin the surprise.

Okay. Here's the surprise. $K_{3,3}$ is also not planar, and has no subdivision of K_5. But that is essentially it. Kuratowski showed the following.

Theorem 11.35 (Kuratowski's Theorem)**.** *A graph is planar if and only if it contains no subdivision of K_5 or $K_{3,3}$ as a subgraph.*

We won't prove this, as it is a little involved. But we should show the following.

Problem 11.28. Show that $K_{3,3}$ is not planar.

One can do this with ad-hoc arguments. But we can also do it more methodically.

11.5.3 Euler's Formula

A planar embedding of a graph G defines *faces* in the plane: the little bits of plane you get by cutting along all the edges of the embedded graph. Where $|V(G)| = n$ and $|E(G)| = m$ and f is the number of faces. Euler showed the following fundamental identity.

Theorem 11.36 (Euler's Formula)**.** *For any embedding of a connected graph G in the plane,*

$$n - m + f = 2.$$

Proof. The proof is by induction on m. As G is connected, $m \geq n - 1$. If $m = n - 1$ then G is a tree, so there is only one face, and the identity holds. This is our base case.

Now assume that $m \geq n$. Then there is an non-cut edge e. Remove it. Its removal joins two faces, so reduces f by one, but also reduces e by one. So the identity holds from the induction hypothesis. □

Corollary 11.37. *If G is connected and planar and $n \geq 3$, then*

$$m \leq 3n - 6.$$

Proof. Count the number N of pairs (e, r) where e is an edge, and r is a face with e on the boundary. As every edge is in at most 2 faces, we have that $N < 2m$ and as every face has at least 3 edges, $3f < N$ so

$$f \leq N/3 \leq \frac{2}{3}m.$$

Plugging this into $n - m + f = 2$ gives that

$$n - m/3 = n - m + \frac{2}{3}m \geq 2$$

so $m \leq 3n - 6$ as needed. □

Problem 11.29. Use this corollary to show that K_5 and $K_{3,3}$ are not planar.

Problem 11.30. ?? Show that any planar graph has a vertex of degree at most 5.

11.5.4 Dual Graphs

Definition 11.38. Given an embedding of a graph G we construct a new (multi-)graph G^*, called the *dual of G,* from G as follows.

The vertices of G^* are

$$V(G^*) = \{v_f \mid f \text{ is a face of } G\ \}.$$

The edges are defined as follows. For every edge e of G let $e^* = v_f v_g$, where f and g are the faces of G incident to e. Then $E(G^*)$ is the **multiset**

$$E(G^*) = \{e^* \mid e \in E(G)\}.$$

(Note that both $V(G^*)$ and $E(G^*)$ depend on the particular embedding we chose for G.)

Not only is G^* a multigraph, it can have loops:

Problem 11.31. What is the dual of a tree on n vertices.

Problem 11.32. Find conditions on a graph G so that its dual G^* has no loops. Find conditions on G so that its dual, under any embedding, has no multiedges.

Problem 11.33. Give an embedding of K_4 and construct its dual. Can you find an embedding of K_4 that gives a different dual? Can you find some graph G with two different embeddings that yield different duals?

Problem 11.34. Show that the dual of an embedded graph is planar.

Theorem 11.39. *Let G be planar and G^* the dual of an embedding of G. Then G is bipartite if and only if G^* is eulerian.*

Proof. One way is easy. Let G be a bipartite graph embedded in the plane. For any vertex v_f of G^*, the walk around the boundary of f is a closed walk. As G is bipartite, it is even, so v_f has even degree. This was for any vertex of G^*, so G^* is euleriain.

On the other hand, assume that G^* is eulerian. We want to show that every cycle of G is even. (This is clear of the boundary of faces under a given embedding but not of every cycle.) Fix an embedding of G.

Let C be a cycle in G, and let $A \subset F(G)$ be the set of faces inside C relative to the embedding. Then C is the symmetric difference of the edge sets of faces in A. Each face has a even number of edges, so so does their symmetric difference. Thus C is even. □

Problem 11.35. Draw an embedding of Q_3. Draw its dual.

11.6 Hamilton Paths and Cycles

Definition 11.40. A path, or cycle, in G is *hamilton* if it is bijective. A graph G is *hamiltonian* if it contains a hamiltonian cycle.

There were some nice conditions that characterised the eulerian graphs. We are not so lucky with hamilton graphs. In fact, the problem of deciding if a graph on n vertices is hamiltonian, is NP-complete. This means the best known algorithm we have for deciding, takes time that is exponential in in n, (at least for some graphs). This doesn't mean we can't say anything about graphs that are hamiltonian. Indeed they must be connected, and have no leaves. We can say a bit more.

Problem 11.36. Find a hamilton path in Q_n.

Problem 11.37. How many hamilton cycles are there in the n-wheel W_n. The W_n is the n-cycle C_n with one more vertex h adjacent to all other vertices.

Here is an early result of Dirac.

Theorem 11.41 (Dirac 1952)**.** *A graph G with n vertices is hamiltonian if $\delta(G) \geq n/2$.*

Instead of proving this we prove a slightly stronger result of Ore.

Theorem 11.42 (Ore 1960)**.** *A graph G with $n \geq 3$ vertices if hamiltonian if for every two vertices $x \neq y$ with $x \not\sim y$, we have $\deg(x) + \deg(y) \geq n$.*

Proof. Towards contradiction, let G be a maximal counterexample on $n \geq 3$ vertices. So G is not hamiltonian, and satisfies

$$(*) \qquad x \not\sim y \Rightarrow \deg(x) + \deg(y) \geq n,$$

but $G + xy$ is hamiltonian for any $x \not\sim y$. (We may assume that G is not complete, as complete graphs are clearly hamiltonian.) Let $H = G + xy$ for some $x \not\sim y$ in G. Then H is hamiltonian, and any Hamilton cycle contains xy. So there is a hamilton path

$$x = x_1 \sim x_2 \sim \ldots x_n = y.$$

Now if $x_1 \sim x_i$ and $x_{i-1} \sim x_n$, then

$$x_1 \sim x_2 \sim \cdots \sim x_{i-1} \sim x_n \sim x_{n-1} \sim \cdots \sim x_i \sim x_1$$

is a hamilton cycle not using xy so in G, which is a contraction. So for each of x_1s $\deg(x_1)$ neighbours x_i in $\{x_2, \ldots, x_{n-1}\}$ (recall that $x_1 \not\sim x_n$), x_{i-1} is not a neighbour of x_n. So

$$\deg(x) + \deg(y) = \deg(x_1) + \deg(x_n) \leq \deg(x_1) + (n - 1 - \deg(x_1)) \leq n - 1$$

This contradicts $(*)$, so the theorem is true.

□

Problem 11.38. Prove Diracs theorem using Ore's theorem.

Problem 11.39. Do problems 10, 18, 21 from Section 11.5 of the text.

11.7 Graph Colouring

Definition 11.43. A *(proper) k-colouring* of a graph G is a function $f : V(G) \to [k]$ such that

$$u \sim v \Rightarrow f(u) \neq f(u).$$

If G has a k-colouring, it is *k-colourable*. The *chromatic number* $\chi(G)$ of G is the minimum k for which G is k-colourable. If $k = \chi(G)$, we also say that G is *k-chromatic*.

Problem 11.40. For what graph H is the following statement true? A graph G is k colourable if and only if $G \to H$.

Problem 11.41. What is the chromatic number of:

i. K_n

ii. C_n

iii. Q_n

iv. W_n

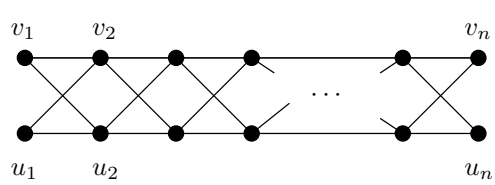
v. The petersen graph.

Problem 11.42. Describe all 2-colourable graphs. (That is, complete the statement " A graph is 2-colourable if and only if ... " using properties we've seen before.)

Problem 11.43. (This is hard.) Let $G(n)$ be the graph (pictured below) with vertices $V(G) = \{v_1, \ldots, v_n\} \cup \{u_1, \ldots, u_n\}$ and the following edges.

- $v_i \sim v_j$ if $|i - j| = 1$.
- $u_i \sim v_j$ if $|i - j| = 1$.
- $u_i \sim u_j$ if $|i - i| = 1$.

How many 3-colourings does the following graph G have? (A recursive answer is fine.)



Definition 11.44. The *clique number* $\alpha(G)$ of a graph G is maximum n such that K_n is a subgraph of G.

Problem 11.44. Show that $\chi(G) \geq |V(G)|/\alpha(\overline{G})$.

Definition 11.45. A graph G is k-critical if it is k-chromatic, but any proper subgraph is $(k-1)$-colourable.

Problem 11.45. Show that K_k is k-critical.

Problem 11.46. Show that any k-chromatic graph contains a k-critical graph as a subgraph.

Problem 11.47. Find all 3-critical graphs.

Problem 11.48. Show that any n-critical graph G has $\delta(G) \geq n - 1$.

From the last two problems and Problem **??** you should be able to do the following.

Problem 11.49. Show that any planar graph is 6-colourable.

In fact, more is true:

Theorem 11.46. *(The Four-Colour Theorem [Haken-Appel 1976]) Any planar graph is* 4*-colourable.*

The proof of this is very long and difficuly. But it is an interesting exercise (maybe also a bit difficult) to prove that every planar graph is 5-colourable.

12 More Graph Theory